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Bachelor Thesis 2026

Organisational commoning

Shelf storage and sharing-spaces



Author: Mio Leuchovius
Supervisor: Mathilda Tham, Miguel Salinas, Juliana Restrepo, Lena Håkansson, Leaha Ireland
Examiner: Matilda Tham
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Abstract

A sharing space as defined within this document is a space, place, or temporary event that facilitates peer-to-peer sharing of second-hand items. Sharing spaces like second-hand stores and flea markets seem to have their design worked out. But little has been written about the category of small grassroots self-service sharing spaces. Yet within these spaces, the design can play a very large role in how the sharing processes function. Institutional retail shelf design exists within a fundamentally divergent operational environment and with different goals and challenges. The free-use nature of a sharing space leads to the arrangement of goods being decided by end users. Using techniques gleaned from firsthand participatory and observational knowledge of sharing spaces, alongside traditional design practises gleaned from academia, this thesis project aims to create guidelines and principles to help gather items into easily manageable, accessible arrangements. The aim is to decrease clutter and reduce cleaning and maintenance time. This will make it easier to create and maintain this type of common sharing space and thereby help create a more circular economy, focused on reuse.

This Design + Change project answers the research question: “How can the development of an organizing and storage system help create easily manageable self-service sharing spaces for a more circular economy?” as well as the sub-question: “How can sharing spaces become more efficient and user-friendly through their design, combatting the tragedy of the commons within the context of Linnaeus University?”

Keywords: Storage-design, Commoning, Second-hand, Circular-economy

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I also want to thank the members of the original RePlace group from last year who were part of getting this project started.

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1 . Introduction

There is a lack of economies that reach a Sustainable Development Goal when it comes to circular economy and reuse (‘Circularity Gap Report’ 2025). To achieve a more circular economy, materials we have already extracted need to be reused, and the lifecycles of already produced objects need to be extended (‘Circularity Gap Report’ 2025). A way of doing this is creating spaces for peer-to-peer sharing of second-hand objects. This allows people to gather the things they need from a network of other people instead of seeking out virgin products. This author has personally been involved in free-use of common sharing spaces for a substantial amount of time. Based on these experiences, the successes of such endeavors, as well as arising problems that can occur and their impact on such operations as a whole, are largely based on the storage system. A novel challenge that can occur within this type of space is related to the shelf design. Within this type of sharing space, end users take and leave items on shelves. Since the user is responsible for placing and thereby arranging the items themselves within the space, the users become collectively responsible for determining the overall organisation of the sharing space. This, in turn, can lead to the organisation quickly becoming unstructured and cluttered, this problem can compound as new user follow the examples of old ones. People managing the space have, as an extension of this, to clean and maintain the space more intensively and frequently, making the space effectively too complex to control.

This independent project is primarily based on the sharing space formerly placed within the facilities of Linnaeus University Campus Växjö, known as “Återbruks Loungen”. It focuses on the complexity of sharing spaces, aiming to combat organisational issues through the redesign of their storage systems, as well as looking at potential scenarios and use cases that this system could fit within. And applying those learnings to potential new sharing types of sharing spaces, as well as the particular sharing space known as the RePlace.

1.1 Terminology

A *sharing space* is as defined within this document a space, a temporary event, or an installation that facilitates sharing by creating a space where people can leave and take items that they no longer need. Examples of sharing spaces could include things like flea markets and second-hand stores. Spaces like libraries would not be considered sharing spaces, though they do promote sharing; they do not facilitate reuse and extend product lifecycles by creating a space for people to leave and take items they no longer need.

Återbruks loungen, or as it was sometimes colloquially known, “ The Sharing Container”, is a free-use open sharing space based out of a container that formerly mostly operated from the Linnaeus University campus. It operated until the end of the spring semester 2025.

RePlace is a sharing space within the Linnaeus University campus, it is the successor project to Återbruks loungen.

Community library is, as defined within this document, a name for small local neighborhood sharing spaces that focus primarily on book exchange. They are most often small boxes put up in places like private gardens facing the sidewalk, or on street corners. Also interchangeably referred to as “small-scale modular sharing spaces”.

Open versus *Closed* is a concept created within this document that articulates how free form any particular type of storage is. It expresses how specialized any particular storage method is and to what degree any random item could fit in it. Something like a close hanger is quite closed, as not much else besides closed can fit on it. Something like a freeform shelf is quite open, as most things can fit on it.

Collectability is a concept created within this document that describes how likely someone is to get more than one example of a category of item. Books are very collectible, as having one does not make you less likely to get another. Blenders are not very collectable; if you have one that functions well, you are very unlikely to get another one.

Consumability is a concept created within this document that describes the degree to which different categories of items get ‘used up’. Books are very consumable once you have read one, you are less likely to read it again. Clothes are consumable, they both break but also go ‘out of fashion‘, which imparts an impulse to get new ones. Frying pans are not very consumable; they do not become less desirable or used up.

2 Theoretical introduction

Commoning is a framework of resource distribution in which resources are not singularly controlled or managed, but are instead treated as community common goods that are managed, controlled, and enjoyed by a community (Gibson-Graham 2013). Examples of successful commons include the “Creative commons” copyright licence (Creat. Commons, n.d.), cases of community-led housing (Arbell 2023), or forestry commons (‘Asociación Forestal de Soria’, n.d.)

The type of sharing spaces addressed within this thesis are heavily related to commoning. During my experiences within the management team of Återbruks loungen, we did not explicitly see ourselves as existing within a common framework. The project was successful in creating a lively sharing environment, but also encountered problems, particularly related to cleanliness and the design of the space. These failures of Återbruks loungen might be characterised as an example of the *Tragedy Of The Commons*. The idea behind the *Tragedies Of The Commons* was introduced by Garrett Hardin in 1968, based on earlier writings by William Forster Lloyd from 1833. Within his article "The Tragedy of the Commons" (Hardin 1968), Hardin tries to explain his view of the futility of the commons, and how they will inevitably devolve.

"The tragedy of the commons develops in this way. Picture a pasture open to all. It is to be expected that each herdsman will try to keep as many cattle as possible on the commons … As a rational being, each herdsman seeks to maximize his gain. Explicitly or implicitly, he asks, " What is the utility *to me* of adding one more animal to my herd?" … the rational herdsman concludes that the only sensible course for him to pursue is to add another animal to his herd. And another; and another . . . . But this is the conclusion reached by cash and every rational herdsman sharing a commons. Therein is the tragedy. Each man is locked into a system that compels him to increase his herd without limit— in a world that is limited." (Hardin 1968)

Hardin believes that cold self-interest will inevitably kill the commons in a large complex society, and they should therefore either be regulated by the government or be sold off as private property.

> "The tragedy of the commons as a food basket is averted by private property, or something formally like it. But the air and water cannot readily be fenced, and so the tragedy of the commons as a cesspool must be prevented by different means, by coercive laws or taxing devices". (Hardin 1968)

> "Equal rights to the commons is to lock the world into a tragic course of action." (Hardin 1968)

"The tragedy of the commons" can occur, and some projects might experience elements of it. However, many scholars, economists, and activists refute its point and showcase how commons can survive.

> "Hardin's famous sheepherder allegory failed to

> make two key conceptual distinctions: the allegory conflated the idea of a scarce resource with the governance of that resource, and it further conflated open-access with commons, despite significant differences " (Frischmann et al. 2019)

Other researchers, like Nobel Prize laureate Elinor Ostrom, have spent large parts of their careers looking at commons and the neoliberal consequences of Hardin's argument.

> "The starkness of Hardin's original statement has been used by many scholars and policy-makers to rationalize central government control of all common-pool resources[3] and to paint a disempowering, pessimistic vision of the human prospect.[4] Users are pictured as trapped in a situation they cannot change. Thus, it is argued that solutions must be imposed on users by external authorities. Although tragedies have undoubtedly occurred, it is also obvious that for thousands of years people have self-organized to manage common-pool resources,"(Ostrom et al. 2017)

Books similar to "Take Back the Economy"(Gibson-Graham 2013) have tried to adapt some of her ideas regarding how the commons remain successful. Take back the economy, in particular, has a section where author Gibson-Graham discusses what separates the functional commons from devolving commons. The book proceeds to set up the following collection of points detailing the guidelines that commons should follow to continue to be successful:

> "• access to property must be shared and wide,
>
> • use of property must be negotiated by a community,
>
> • benefit from property must be distributed to the community and possibly beyond,
>
> • care for property must be performed by community members,

and

• responsibility for property must be assumed by community members.“

(Gibson-Graham 2013)

During this project, a particular focus lies on the points “access to property must be shared and wide,” and “care for property must be performed by community members”. When looking back on the example of Återbruks loungen, it is evident that the shelf design and organisation systems have kept these points from being fulfilled. While that may not be the sole contributor to the failures of Återbruks loungen, it is definitely a significant factor. The shelving system was designed in a way that made cleaning and organizing difficult. This factor makes the shared caring for and by community members harder to achieve, even if community members want to contribute to its cleanliness. The unkempt appearance also might have acted as an inhibitor towards Gibson-Graham’s first point, the mess acting in and of itself as a challenge towards open-access, as people would not feel invited into such a space. It also might have made people view Återbruks loungen more negatively, which I speculate might have made them less willing to help care for the property, and lowered their understanding of the Återbruks loungen as a place that could be kept clean.

The goal of this Project consists of looking at the storage design of free-use open-access common sharing spaces and designing them to help promote order and minimize maintenance. This guides the sharing space projects looked at within this report towards “take back the economy" 's commoning point, “ access to property must be shared and wide,” and “care for property must be performed by community members”. Thereby, helping avoid the “Tragedy of Commons” and allowing common sharing spaces to prefiguratively help move us further towards a circular economy.

This can further be exemplified and visualized through the meta design nest, a design tool used within the design design+change program at Linnaeus University. (Lundebye 2004) (Tham 2014) (Tham et al. 2019), c

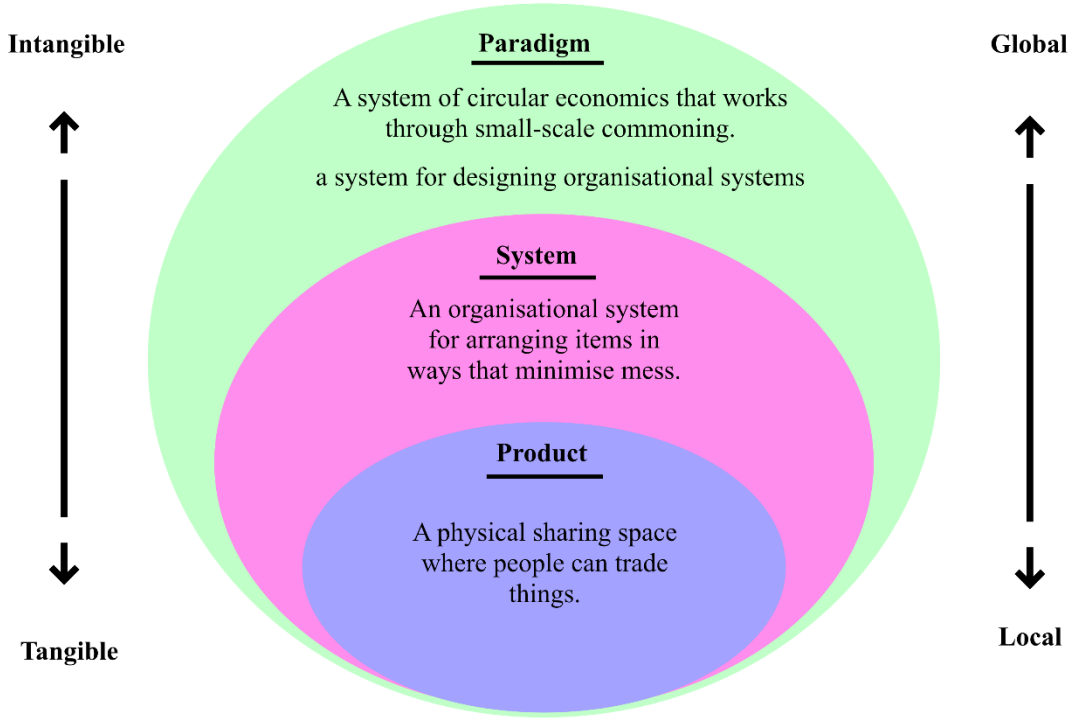


Figure 1: Metadesign map

3 Methodology

Among the methods I used, one of the most impactful ones was that of case studies. In practice, this meant that several different examples were analyzed and dissected to learn how they functioned and what the problems were within them. Major examples of this were a case study of Återbuksloungen, community libraries, and retail shelf layouts.

When doing case studies, I tried to gain multiple types of data in each case, though the type of data in each was different. Examples of this might include researching traditional retail design theory whilst also gathering photographic data and experiential knowledge of interacting with the

system. Återbruks loungen, on the other hand, consisted of long-term observational data gathering on usage patterns.

Another method used was collaborative synthesis. I am just one actor within a larger system. The sharing space projects that this project revolves around are largely interconnected and have many actors. So a continual part of the process has been talking with these other actors and synthesizing their views and understandings of the systems with my own to create a more complex understanding of the systems.

Scenario development has also been a powerful method for this project. The goal of this project is to explore multiple possible sharing spaces to create new methods and tools for designing them. However, creating multiple physical materialisations in real life takes more time than would be possible within this project, so instead, scenarios have been developed.

4 Design

4.1 Återbruks loungen

This project originates from working with the Återbruks loungen and seeing the kinds of problems that the lack of a properly thought-out design could cause for this type of space. Using Återbruks loungen as a case study and comparing its design, problems, and successes with those of a traditional store layout, we can gain an understanding of how they differ and how to change the design.



Figur 2: Återbruks loungen

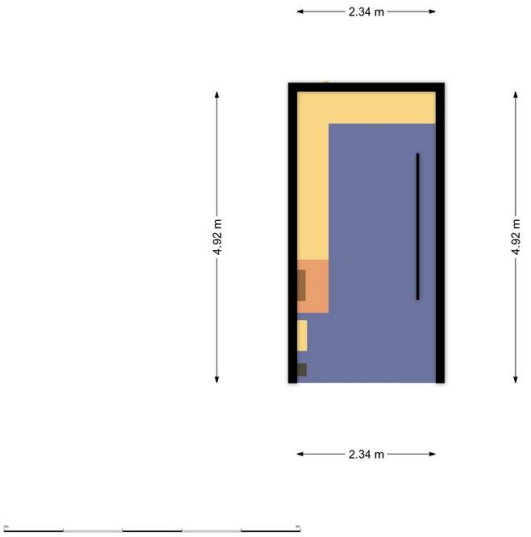


Figure 3: Återbruks loungen layout

The project started with a retrospective analysis of the Återbruks loungen and its difficulties. In opposition to sharing spaces, how to design shelving for retail spaces seems to be a relatively developed problem. Papers like “Joint Shelf Design and Space Planning Problem With Placement Options” (Zajac and Kuhn 2025) and ”Joint Shelf Design and Shelf Space Allocation Problem for Retailers” (Gecili and Parikh 2022) focus specifically on the area of interest for this project, that is, shelves and how to design them. However, after researching shelf design and retail space management, it became increasingly clear that the problems in this field are significantly different from the problems present within free-use sharing spaces like the sharing Återbruks loungen.

The field of retail design seems to be primarily focused on how to increase the “visual representation of products on shelves”(Gecili and Parikh 2022) and ways to “generate a profit-maximizing shelf layout” (Zajac and Kuhn 2025). While the sharing space problems are of an organisational nature, like creating a clear system for end users to easily sort their goods into.

The analysis of these problems stems from observations and experiences of being responsible for the management of Återbruks loungen for one semester. Further, I based my analysis on the input of other members of Återbruks loungens management team in the student association Klimatstudenterna. While working on designing an alternate spatial layout for Återbruks loungens, I conducted several excursions and field studies to examine the design, layout, and shelf management of various retail spaces and second-hand stores.

The retail spaces examined consist mostly of flat, height-adjustable shelves with pegboards or a hanging display.



Figure 5: Field reserch Åhlens, Växjö



Figure 6: Field reserch Åhlens, Växjö



Figure 8: Field reserch willys, Växjö



Figure 4: Field reserch willys, Växjö



Figure 7: Field reserch Emmaus, växjö

This design is assumedly optimal for an effective visual representation of products, but it is not ideal for a self-service sharing space like Återbruks loungen was.

Using Återbruks loungen as a case study, I want to look at how this type of design falters in the challenges presented by free-form sharing spaces, like Återbruks loungen, or its successor project, the RePlace. The RePlace and Återbruks loungen have a different goal than traditional retail spaces and therefore require different solutions. The fundamental difference between retail spaces and sharing spaces is the free-use nature of them. Retail spaces are managed and organised by employees and retail staff; they are the ones deciding where items should go and restocking them. Within free-use sharing spaces, the organisation is instead managed impromptu by the end user. Återbruks loungen had a management team that was responsible for cleaning it, but the end users who left objects at Återbruks loungen were the ones placing items on the shelves. The management team at Återbruks loungen could and did rearrange items, but the fundamental shape of how things were organised was shaped by the whims of end users.

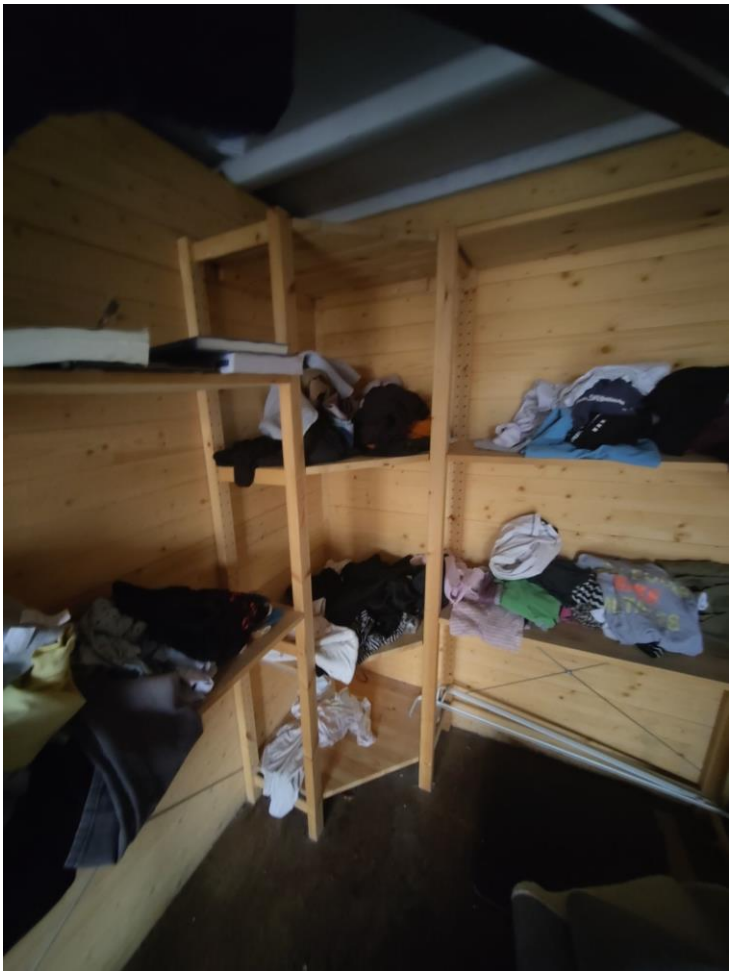


Figure 9: Återbruks loungen ’mess’

It should be noted that "Indifference" was not the major problem affecting Återbruks loungen . Observationally, many users seemed very intent on trying to clean Återbruks loungen. The commoning guideline of “care for property must be performed by community members” didn't fail to be met due to the users being apathetic to its care, but because Återbruks loungen consisted of a storage system that was not adjusted to its specific use.. When a space itself isn't designed to be cleaned easily, it is difficult to keep it tidy.

Some users cared about the state of the sharing space; however, that was not the only problem because of the wide range of users of Återbruks loungen, and its reach went beyond its community. That is another aspect in the commoning guidelines, the “use of property must be negotiated by a community” due to the open-access nature, people beyond the University campus started using it. That resulted in people who did not see themselves as being part of this community and were not as invested in the upkeep of the space.

A particular section of Återbruks loungen, which seemed to work quite well, in opposition to the rest of it, was the clothes hanger section. Looking at this section, one clear difference in the storage medium itself became apparent. Hangers function quite differently from traditional shelves in the way that they are selective with what type of items can be stored with them. On a shelf, most things will fit. On a clothes hanger, however, very few things besides clothes will fit. This particular dynamic I will refer to as *Open* versus *Closed.* This is a tool I developed for looking at how flexible a shelving option is. Storage systems like clothes hangers are a *Closed* system. There is a limited number of items that can be placed in a limited number of ways on it. Something like a freeform shelf is very *open;* most items can be placed on it, which allows you to accommodate most random things users bring, but does also make it easier for the cleanliness to degrade.

Different factors can affect how *Open* or *Closed* something is, beyond the degree to which it physically hinders the user from putting other items there. A separate section that seemed to work well was this impromptu bookshelf that formed on one of the shelves.



Figure 10: Återbruks loungen 'bookshelf'

There was nothing special about this shelf besides the fact that it acted as a bookshelf. It is probably because users already have a pre-existing understanding of how to interact with this

type of interface, also referred to as a mental model (Jones et al. 2011) of what a bookshelf is and how to interact with it. It is obvious to most users that something like a frying pan or a pair of shoes doesn't belong in a bookshelf. That is the second part of the *Open* versus *Closed* dynamic. A person's previous mental model and cultural knowledge of how to interact with spaces also play a large role in how they choose to organize things.

Not every type of shelf or storage method should be *closed*. It is, in fact, very important to have *Open* spaces. A sharing space needs to be able to accommodate all items users want to give, and it is not feasible to construct specialized shelving methods for each distinct type of item. As such, another type of method that can help control mess is adding dividers to shelves. The shelving within Återbruks loungen consisted of several long, uniform planes that ran through most of Återbruks loungen lengthwise. This meant that there was no distinct separation of most areas, which meant that mess could migrate along the shelves. Adding dividers would contain messes more, and would also potentially make it easier for collections of items to form within these sections. Having a non-permanent movable divider would also be an easy-to-use way for the people managing the space to adjust the layout and divisions to different situations.

It is important to make sure that a shelving layout accommodates specific items. An example of this not happening was the section positioned at the far end of Återbruks loungen, where people tended to place clothing. The distance between the shelves was not optimal. This shelving had too much distance between each level. Clothes can only be stacked so high before the stack becomes unstable. But when placing clothes down in Återbruks loungen, people would attempt to stack them up to the shelf height, resulting in unstable stacks of clothing. These would fall over when users attempted to examine clothes lower down in the stack. So, considering specific item requirements that go beyond trying to make a specific space more closed is also important.

4.1.1.1 Analysis

It needs to be noted that shelving design would not solve all of Återbruks loungens problems. It was not always extraordinarily messy, but it functioned well for the most part. Even though it was quite commonly unorganized in most cases, it was not to an extent that would have impacted the accessibility. There was a very consistent flow of items going in and out of the space. Shelf redesign could not have addressed all the problems on it own.

Another example of a sharing space exists in the region of Kalmar Län, and it is reportedly functioning well and has no major problems with organization. The project, which is called TaGe, consists of another container that acts as a place where users can take and leave objects (‘Säg hej till TaGe!’, n.d.). The major operational difference between TaGe andÅterbruks loungen seems to be that TaGe moves around from place to place quite often. Återbruks loungen did this to some extent. During certain weeks or certain times of the year, it would be moved to different places around Växjö. But for the most part, Återbruks loungen had its home on the campus of Linnaeus University. Frequent moves, like in the case of TaGe, might be why it does not get as unorganised to the same extent as Återbruks loungen. Frequent moves might mean

that, as people are more likely to litter in an already unkempt environment (Bergquist et al. 2021), the TaGe container might never stay long enough for that initial treadmill of messiness to start. Most probably, the container gets cleaned before a move, just like Återbruks loungen did. This might mean that TaGe exists in a circumstance that is closer to the "traditional retail environment, as there are dedicated people who clean it quite regularly. It seems like the more organized the cleaning is, the less the particular design matters.

In the case of Återbruks loungen, perhaps having shorter opening hours or having Återbruks loungen only be open on alternating weeks could have allowed us to keep up with the maintenance better.

4.2 Community library

Another type of space that was examined for this project was community libraries.

Community libraries, as defined within this document, are small boxes that individual people build and place around their own local neighborhoods. They serve as small, modular, distributed facilitators for peer-to-peer sharing. When looking at them from the *Closed* versus *Open* perspective, the format is in itself relatively *closed*. The size of the box and the spacing of the shelves limit what types of items could physically fit into the space. Similarly, the mental model of a bookshelf also impacts the types of items that could fit into this type of space. From examining some local neighbourhood libraries on several occasions, it seems that mostly books and adjacent materials like DVDs and CDs are present in community libraries.



Figure 11: Citty 'Neighborhood Library



Figure 12: citty 'Neighborhood Library'

This format is interesting in how hyper-localized and place-dependent it is. The small-scale of it also addresses some of the messiness issues by virtue of a small space being very easy to clean quickly.

When looking if this format could be adapted from the specialized “community library” to a more general, small-scale, modular sharing space, a couple of things appear. The small scale will make it easier to clean, but it will also fit fewer items. Fitting fewer items also means there is going to be less space for *Closed* storage areas. It is also probable that the smaller scale will lead to a more localized user base. Consisting of the people who live directly near it and or whose regular walks pass by it.

The hyper-localized nature of it could likely change the dynamics of how people interact with it. Since it will be more convenient to check several times a day as you walk past.

Since space is such a precious commodity within this format, the question of what kind of items a small-scale modular space should focus on becomes relevant. The goal is to expand beyond the singular category of books, but what else fits here?

The first, most obvious constraint to consider is size. This cannot become a sharing system for large items. That narrows the scope and defines what a small-scale modular sharing space could look like quite significantly. One also has to consider how fast items move in relation to their size. There are items like a microwave that could most likely not physically fit into this kind of space. But there are also several infrequently shared items like plates or forks that would be welcomed in a larger space, but in a smaller area, they become a bigger burden. Infrequently shared items in general become a bigger problem, as noted earlier, it is most probable that a small-scale modular sharing space would have a smaller reach and have users come from a smaller, more localized vicinity. Meaning fewer end users participate in the sharing and taking, and therefore fewer people search for infrequently shared items. It is likely that plates would have a similar effect within small-scale modular sharing spaces as they absorbed large furniture like couches and beds have within Återbruks loungen. They are large, cumbersome items that block space and are difficult to get rid of.

With this in mind, a small-scale modular sharing space should probably not be turned into a generalist sharing space that accepts everything, but instead focus on being more specialized.

Looking at what types of items such a specialized sharing space should focus on, a potential way forward is specializing in items that have been successful in other sharing spaces. Books worked very well within community libraries, clothes seemed to work similarly well for Återbruks loungen.

Examining what these have in common, a couple of factors appear. A major one is the fact that they are both categories of items where users would want to own multiple items of that category.

Owning one piece of clothing is not enough; people want to own several. Similarly, if you own multiple books, that doesn't mean that you would not like another one. Other categories, like e.g., a blender, the owner is very unlikely to want more blenders if they already have one that they like and that works well. This factor of how desirable owning multiple items within one category is, I will within this report refer to as *collectability.*

A second factor I have developed is looking at how *consumable* an item is. Books are not used up, but once they have been read through, it is unlikely they will be immediately re-read. This creates an impulse to look for other books. The impulse to get more clothes does, in part, come from them wearing out, but also in part from certain users seeking out specific clothes that match and express their own unique personal style.

With these two factors in mind, I started developing several scenarios where this type of sharing space could fit, and iterating on potential designs for them.

One of the scenarios was a street corner placement. Such a location doesn't allow for a lot of space so the shelf became quite small. Its target group would be foot traffic and nearby houses.

I imagine objects like trinkets would fit well together with regular items like books.

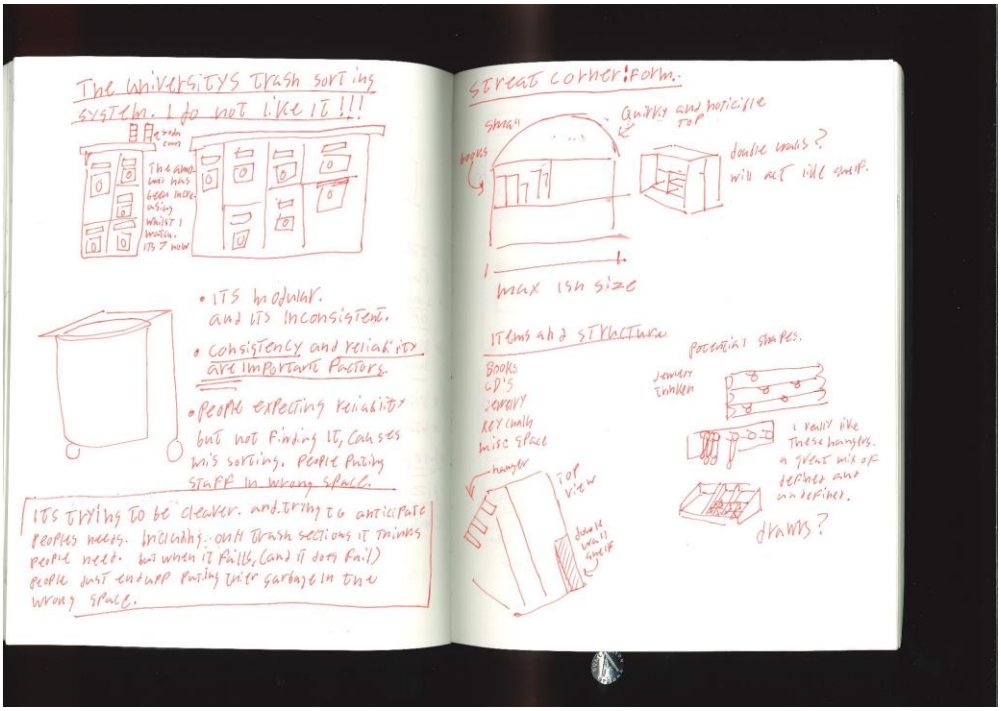


Figure 13: Prooject book sketches

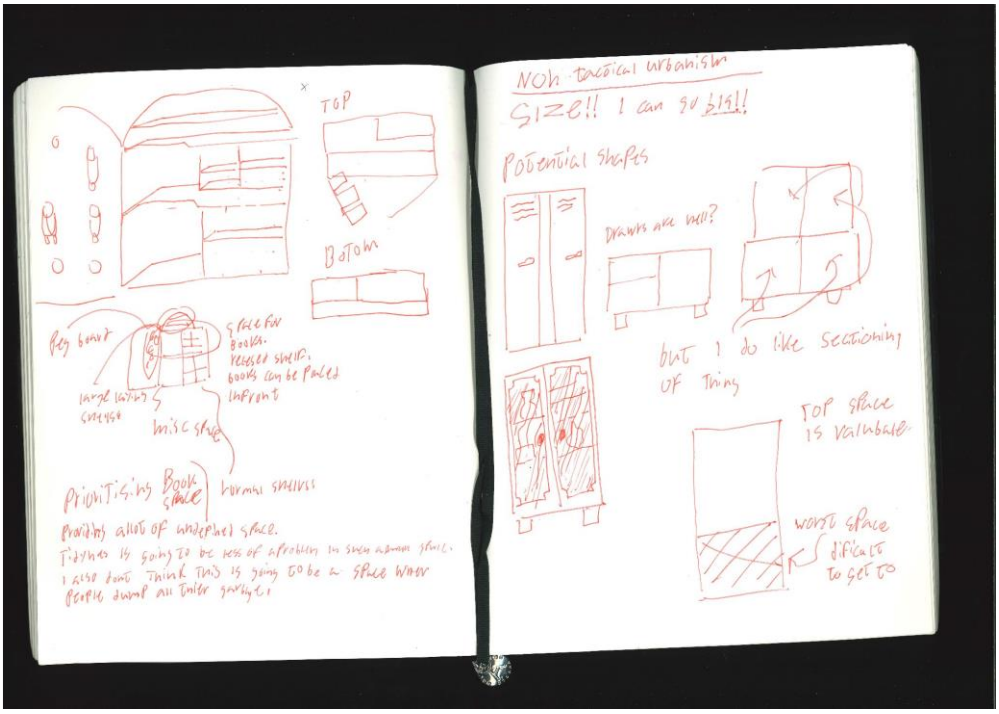


Figure 14: Project book sketches

One of the other scenarios that was developed was that of a bus stop placement. This placement allows for a larger locker-like enclosure. The target group here is similar to the target group of the street corner scenario, but slightly different. I imagine it being placed on a bus stop. Which would allow many people to pass by it, though, of course, primarily people who lived close by. This sharing space would be ideal for the standard things like books. But it might also be large enough to accommodate a small clothes section. Slide-out drawers and shelves would make the entire depth of the shelf accessible. Locating it next to a trash can could help reduce clutter.

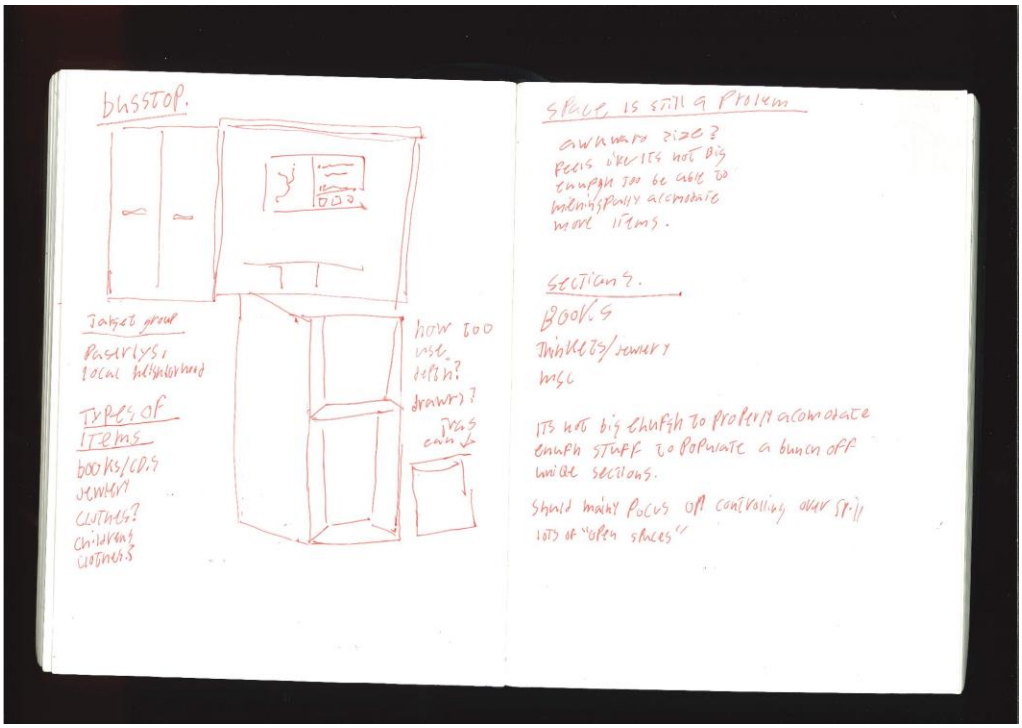


Figure 15: Project book sketches

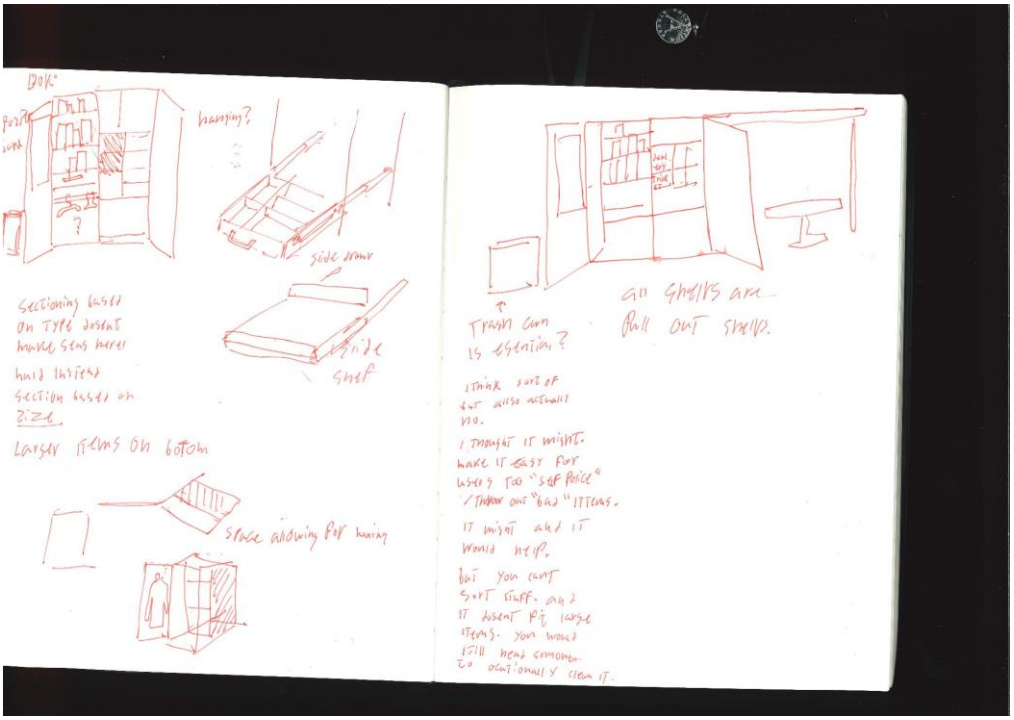


Figure 16: Project book sketches

4.3 RePlace

The third sharing space that was examined for this project was the RePlace, which is the follow-up sharing space project developed by the local University organisation “KlimatStudenterna Växjö.” We were approached by the University, offering us a location inside a permanent building during the time I was working on the redesign of Återbruks loungen. The idea of a place for trading furniture on campus had been discussed many times before, and during that exact time, another group of students interested in such a project was forming. They ended up joining our project, and the new space was given a dual mission of facilitating both the peer-to-peer sharing of items and of furniture

Those students helped out during the first phase of the project, though Most of that initial group left a couple of months later at the end of the term, but the dual purpose of the now named ‘RePlace’ remained. As many other students had left, I assumed the role of primary facilitator for the project. Communicating with different organisations, as well as guiding the development and leading the discussions we had regarding the RePlace within Klimatstudenterna.

The project has had many actors involved. The primary organisations involved were Klimatstudenterna and Linnaeus University, who made the premises available for us to use. The student association “Culture Connections” wer brought in to help with the management of volunteers, and to further secure the project's future. The student union “Linnékåren” was involved with helping us communicate with the University and exerting pressure on the University to enact its previous promises within a reasonable time frame.

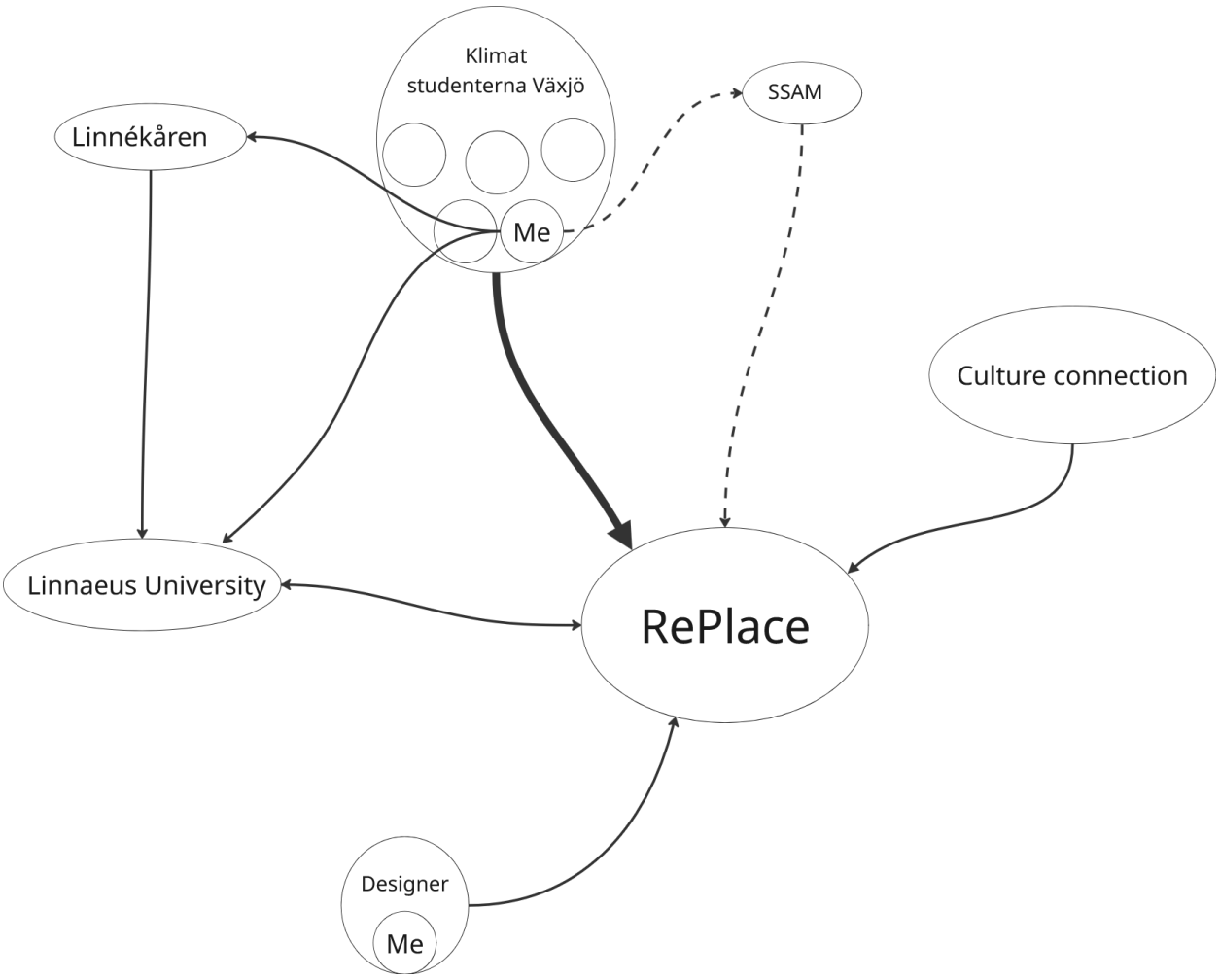


Figure 17: Diagram of RePlace actors

Within the project, I had dual roles. Both as a facilitator and as a designer. My involvement with the project as a designer meant looking at the RePlace and understanding how things could be organized and stored in the most efficient and user-friendly way. It was clear from early on that we were most likely not going to be able to lay out the space to my ideal vision, due to the premises being rented to us by the University, and consequently, the restrictions on how the space can be modified. There is also the second limitation of cost and difficulty in acquiring specific furniture. We were in talks with some organisations, like the municipal trash management company “SSAM” and the local Red Cross second-hand store, to acquire more specific furniture. But nothing concrete came of this. With those limitations in mind, as the lead facilitator working on the physical RePlace project, I was responsible for developing a scenario of how we were going to use the space. But as a designer working on this document, I also decided to expand upon that concept to create a more ideal scenario layout for how we could arrange things if some of our existing limitations were not in place.

The earliest sketches of the RePlace layout were rough spatial drawings that I designed before we even had access to the exact dimensions of the premises.

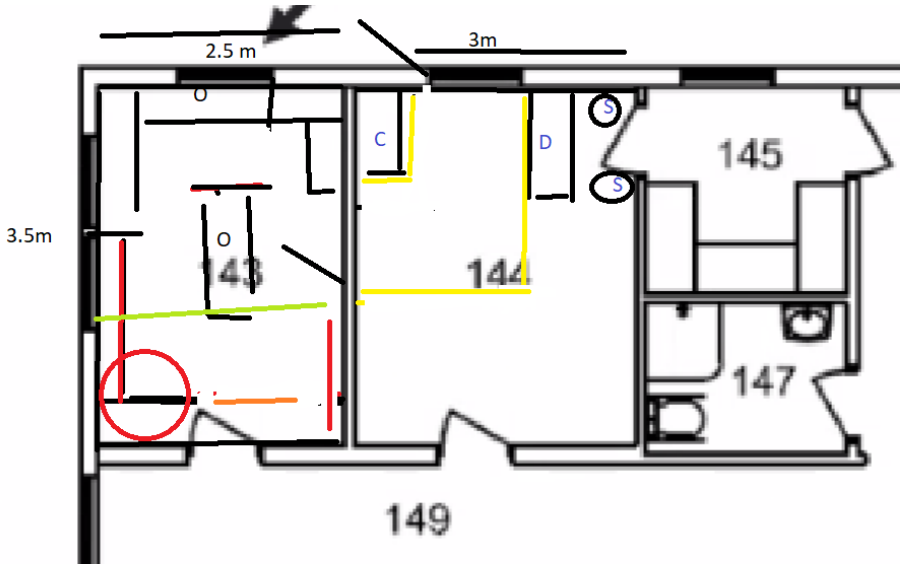


Figure 18: Original RePlace sketch layout

The sketch depicts the early layout. We had gotten two different premises as options to use from the University. But we ultimately chose this one due to its potential expansion options.

This was then cleaned up and made into a concept sketch to show other stakeholders what we were envisioning. With the right room (room 144) serving as the entrance and the left room (room 143) as the second room

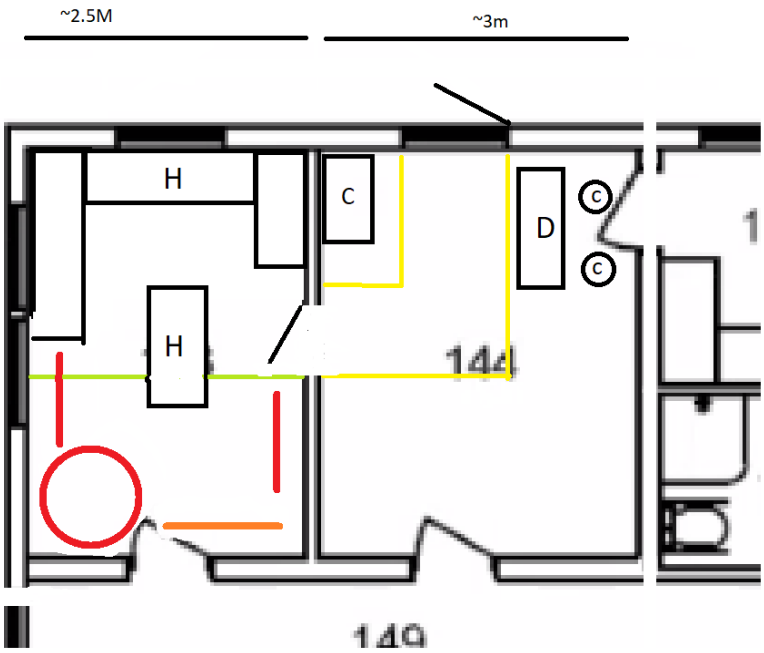


Figure 19: RePlace sketch layout

Here are depicted the first two rooms that would contain the RePlace. Room 144 is where the University would install an entrance door. Moving furniture is cumbersome, especially when it has to pass through multiple doors; as such, I thought it was best to make the first room the furniture's storage space.

Generally, space is very limited, which made us think carefully about which objects were needed and how to arrange things in the best way.

The yellow lines mark the passageway that needs to remain free for users to be able to pass into the second room. The red object in room 143 indicates a clothing stand. The circular clothing stand was placed in the corners to maximize space.

The RePlace is supposed to operate in 2 phases. The first phase was the standard operational phase, where the layout looked like the one pictured above.

The second phase is intended for the end of the semester, when people move out and need to dispose of furniture. The RePlace is supposed to compress as much as possible and enter a phase where normal sharing operations are no longer the goal. The goal is now to store as much furniture as possible inside the space as possible, even if that makes the space otherwise inaccessible.

The green line in the left room, room 143, indicates the space that all its normal phase 1 furniture is supposed to compress into during phase 2, so that more of the user's furniture can fit in that space.

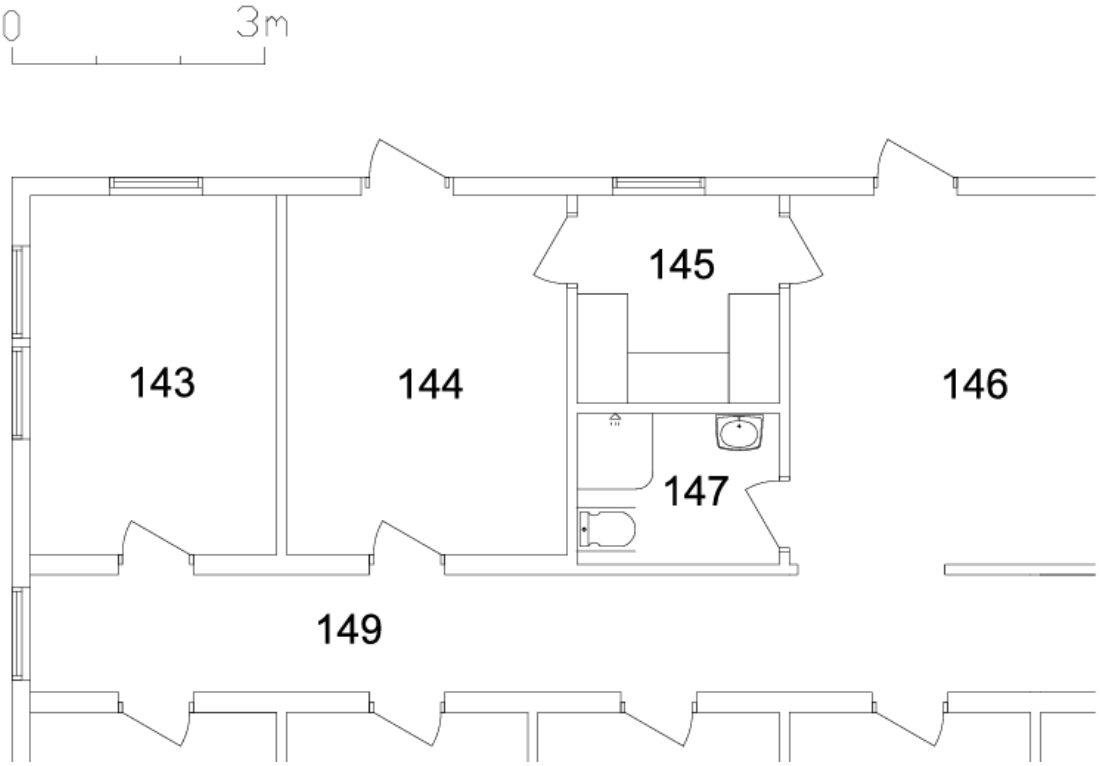


Figure 20(‘Gulan_planritning (1)’, n.d.)

The University eventually sent a new updated plan with their renovation changes

They also decided they did not want to install a door between rooms 144 and 143, which ended up changing parts of the layout.

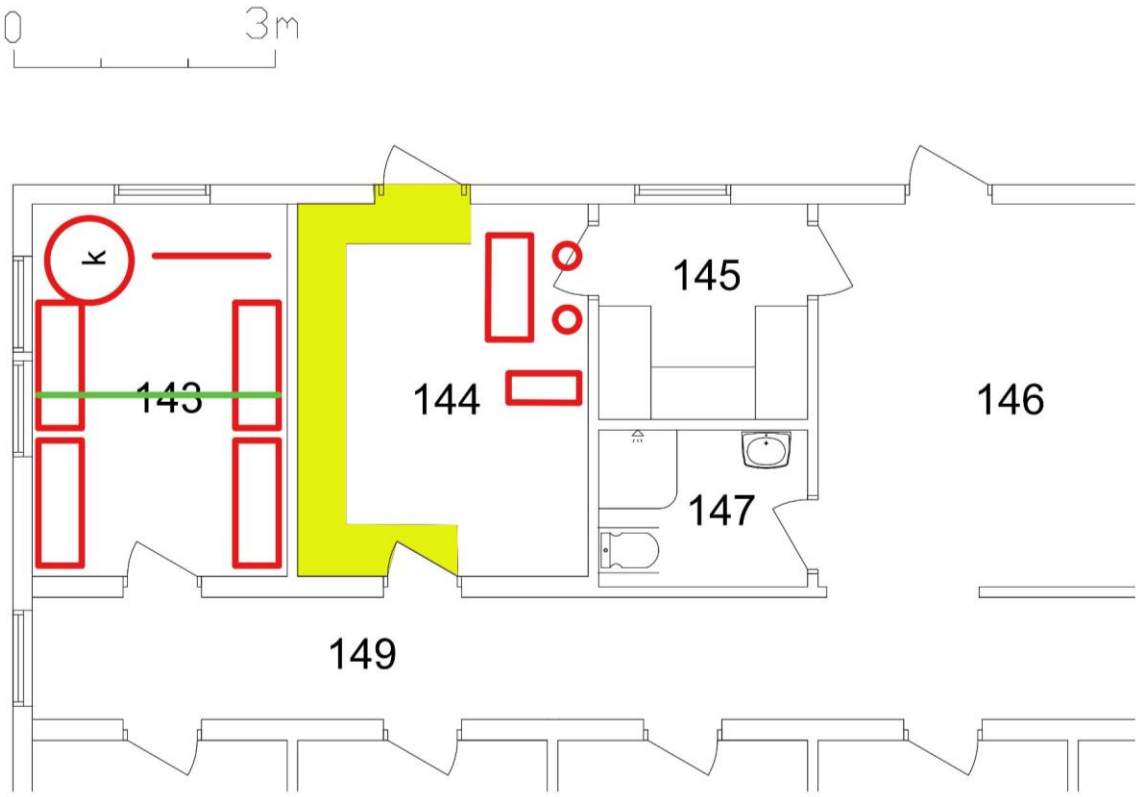


Figure 21: Modified RePlace layout

The previous design was then adapted to the new floor plan. This, to my knowledge, is the rough structure of this design that remains the intended layout for the space, though furniture constraints will apply. The new board of Klimatstudenterna might also decide they prefer a different one.

It was at this point that I decided to create an ideal scenario layout based on the previous one. I decided to start looking at the space as a 3d model to gain knowledge of how the space feels with various furniture in it.

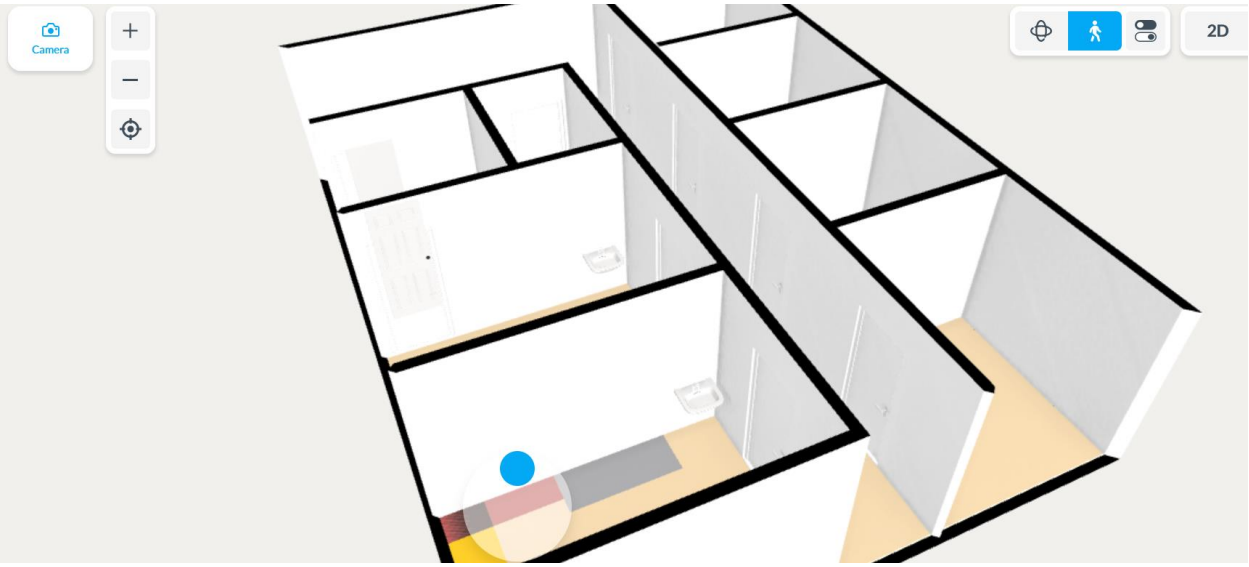


Figure 22:3d model of RePlace layout

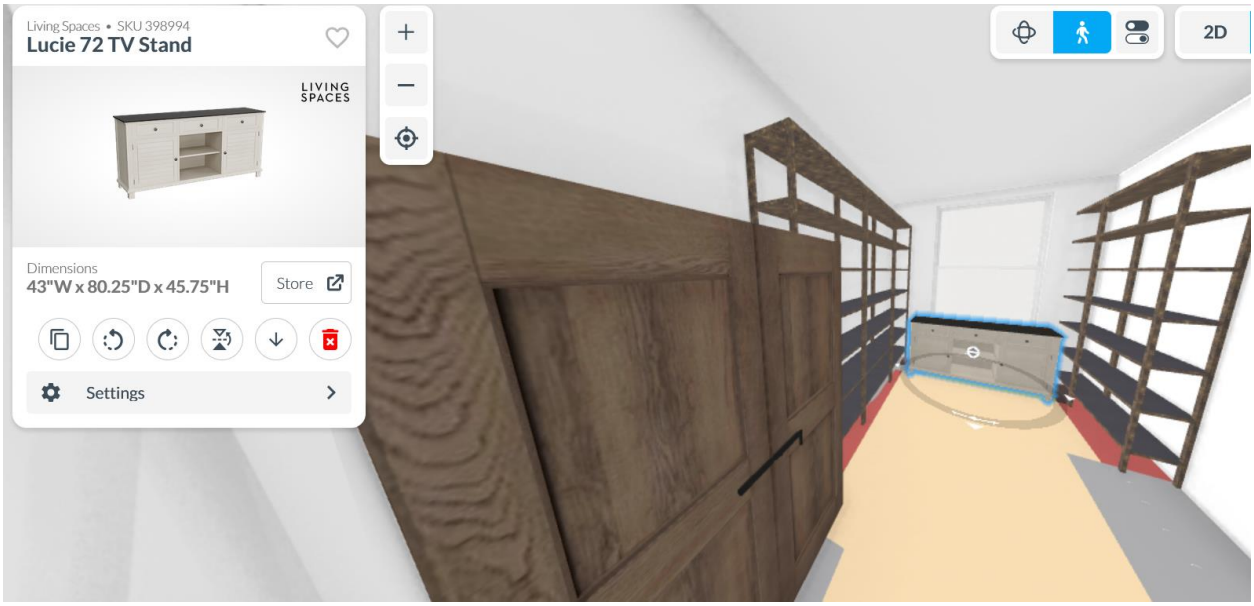


Figure 23:3D Perspective view of RePlace layout.

The model was not used to find the exact furniture that was ideal for the RePlace. The specific furniture within the 3D model were instead added as analogues, to see how the space felt with furniture of different dimensions in various places.

This then led to a round of finalised sketching.

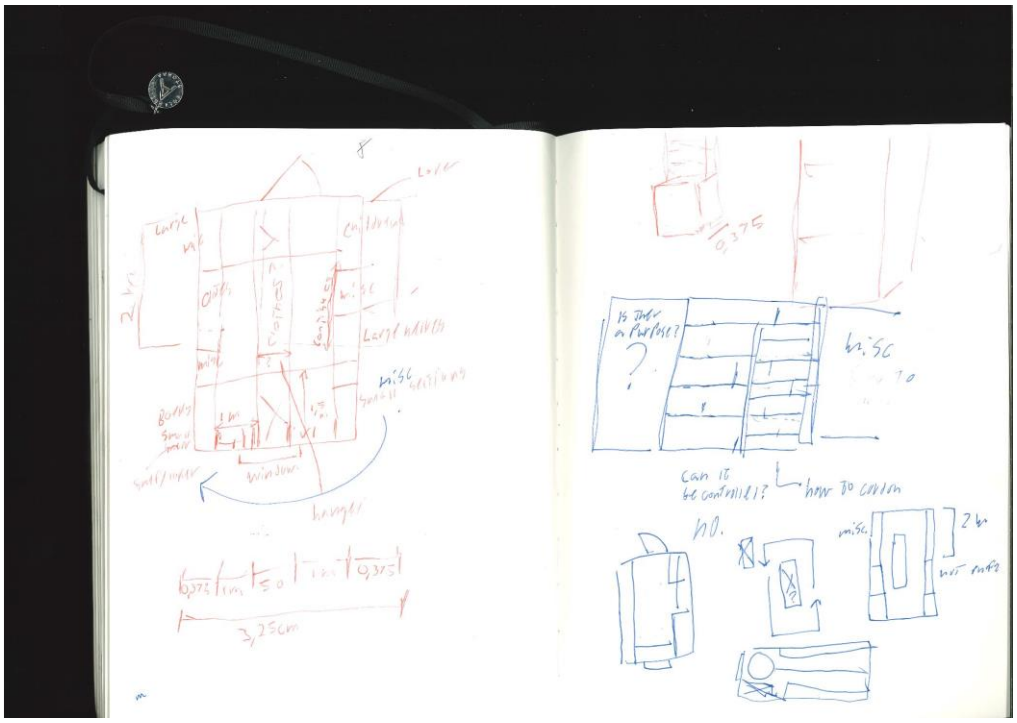


Figure 24: Project book sketch RePlace layout

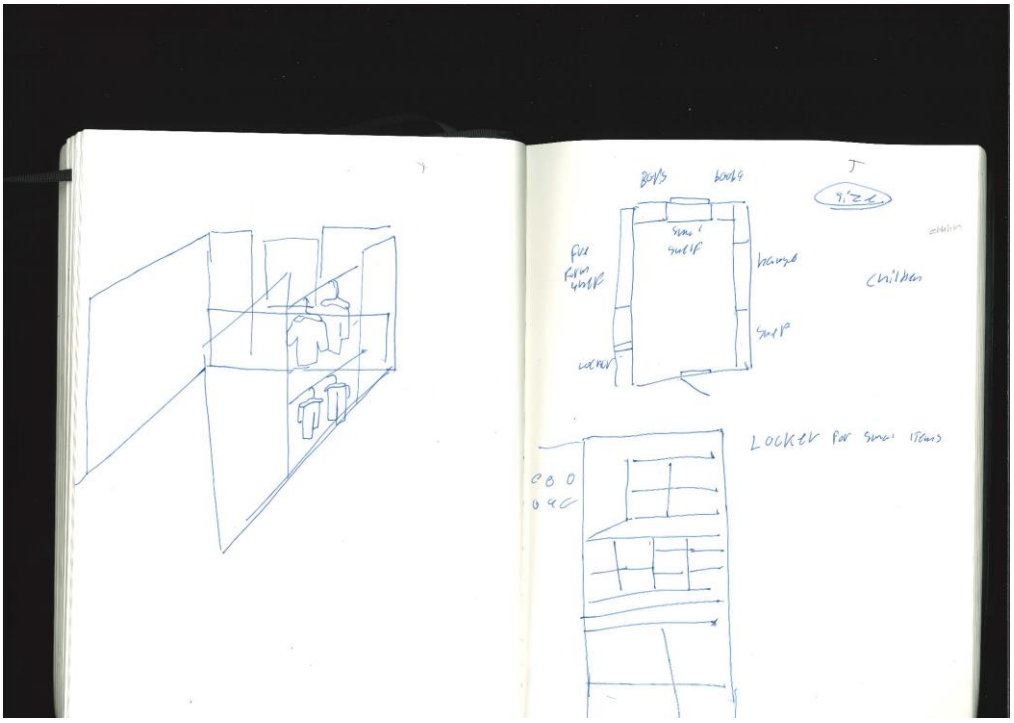


Figure 25:: Project book sketch RePlace layout

From which, in conjunction with all the previous learnings, a finalised scenario layout was developed.

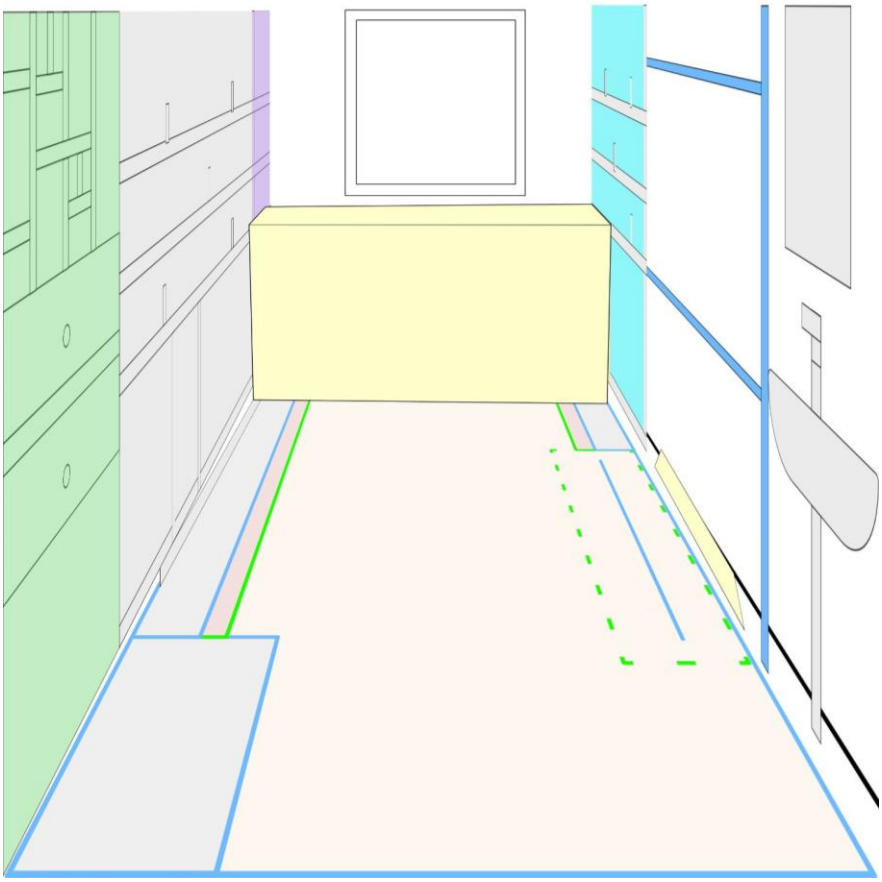


Figure 26: Final scenario layout visualization

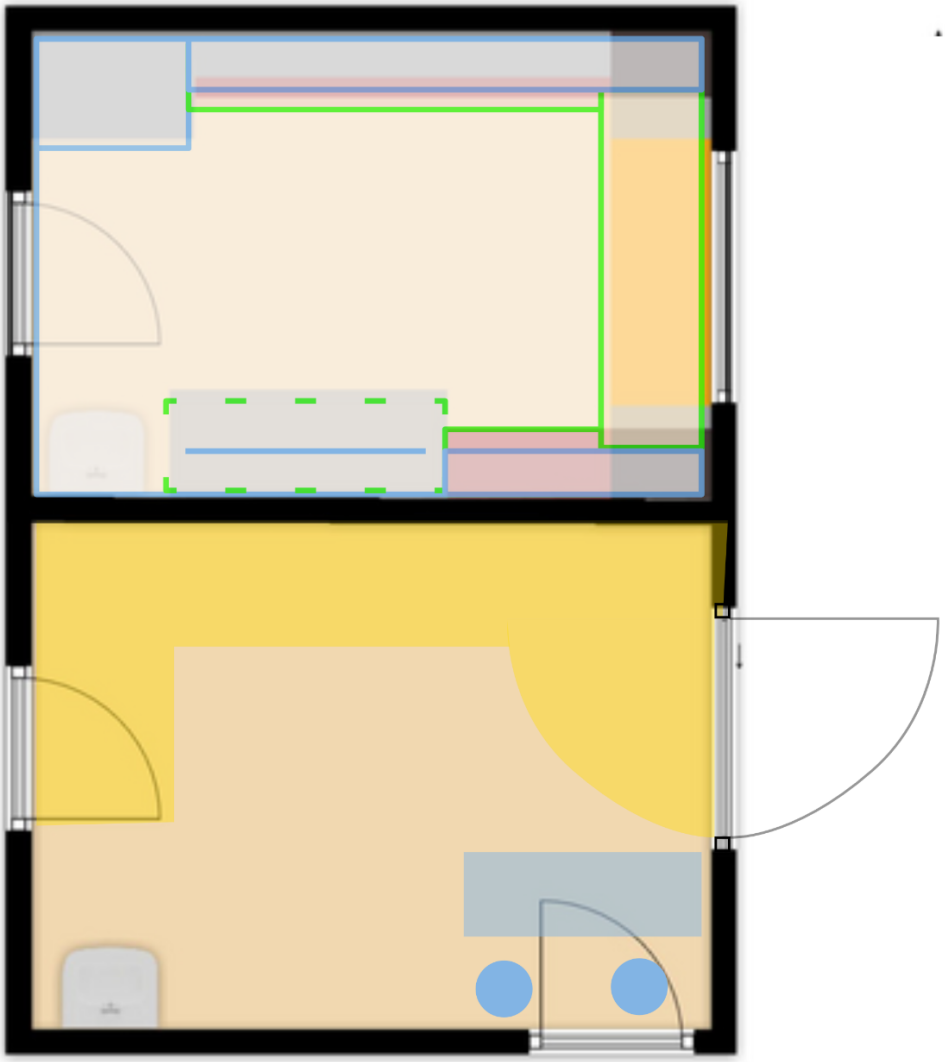


Figure 27:Full final RePlace Layout drawing

5 Outcomes

The sharing space scenario was made and developed along with the learning of this project. This project set out to examine sharing spaces through the lens of commoning and using Take Back the Economy's point for how to run a successful commons (Gibson-Graham 2013), to gain an understanding of how they can be applied to sharing spaces. In an effort to do so, alongside scenarios and visualizations, a list of points of commoning was explored.

5.1 Outcomes: guidelines for sharing spaces

Throughout my learnings within this project, I have reached an understanding of the needs of a sharing space and developed and refined the following nine guidelines for creating a good storage system for a self-service, sharing space.

Here are the 9 guiding points for sharing spaces

(1) Closed Shelving

- Use *Closed* storage types to help define spaces. And make it easier to sort items for users.
- Use *Closed* shelving only for items in categories with a high collectability

(2) Open Shelving

- Make sure to leave enough *Open* freeform shelving space to accommodate the abundance of potential unclear items that might enter.

(3) Mental Models

- Consider users' mental models when designing *Closed* shelving. It can create thematic spaces that make it instinctively obvious to the user how to interact with a space.

(4) Collectability

- Consider *Collectability* when assigning which items get *Closed* shelving spaces.

(5) Separation

- Use dividers to separate spaces, contain messes, and create the possibility for collections to form.

(6) Stackability

- Varying shelf height can act as another way to *close* off a shelf. Specific items may not work well with specific shelf heights. Avoid items being stacked too high by considering which items are supposed to fit on a shelf and which spaces those items end up using.

(7) Visibility

- Use caution when installing drawers, cabinets, or storage items with doors on them. They can act as a means of separating certain types of items or defining a space. But the limited visibility can also make it easy to forget that items inside this type of space exist.

(8) Management

- All sharing spaces are going to need some degree of management; expecting them to be entirely "self-cleaning" is unrealistic. If a pace is accumulating a mess faster than can be managed, consider limiting opening hours to slow that pace, as well as looking at the community managing the space, and whether it would be possible to expand it.

- Determine to what degree the sharing space is self-service versus managed. The more a space is managed, the less a specialized design is needed.

(9) Flexibility

- It is impossible to create an ideal design without user input. Most likely, some part of the shelving system will have to be rearranged after opening. Make sure to build systems flexibly so they can be rearranged.

5.2 Outcomes: scenario layout

Using the 9 points was also integral to designing the finalized scenario layout.

Point (7): Visibility was used in two places. The green area on the right wall represents a locker, and has the aim of acting as an autonomous mini sharing space for specifically smaller items. Separating it visually from the rest of the space would hopefully keep its contents contained. At the end of the room, there is also a yellow drawer designated for kids' clothes. Kids' clothes had a habit within Återbruks loungen of migrating to other spaces, and so it seemed wise to create a distinct space for them.

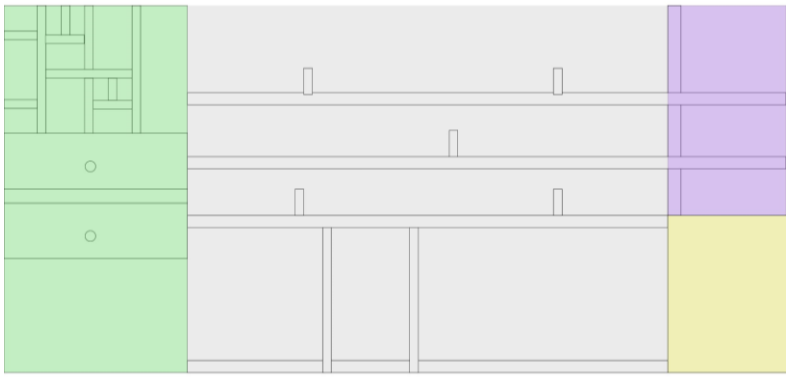


Figure 28: RePlace scenario layout 'left wall'

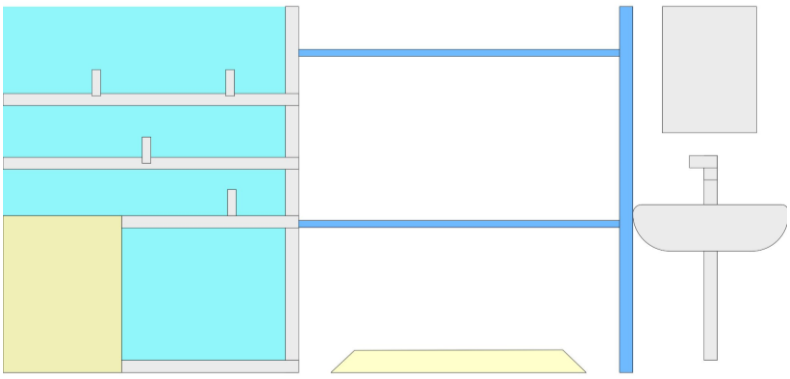


Figure 29:v: RePlace scenario layout 'right wall'

Point (2) Open shelving and point (5) separation are also exemplified in the space. The left wall is mostly open shelving, but it is separated physically from other sections in ways that keep those sorts of items there. There are also plenty of dividers to contain messes.

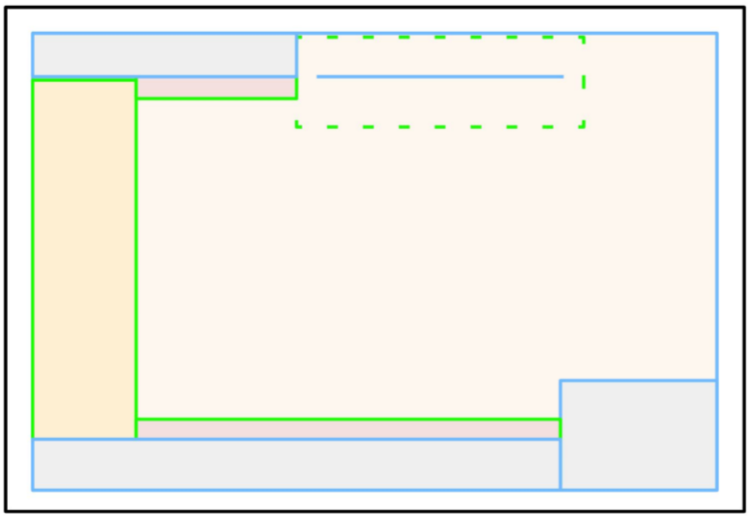


Figure 30:: RePlace scenario layout 'floorplan'

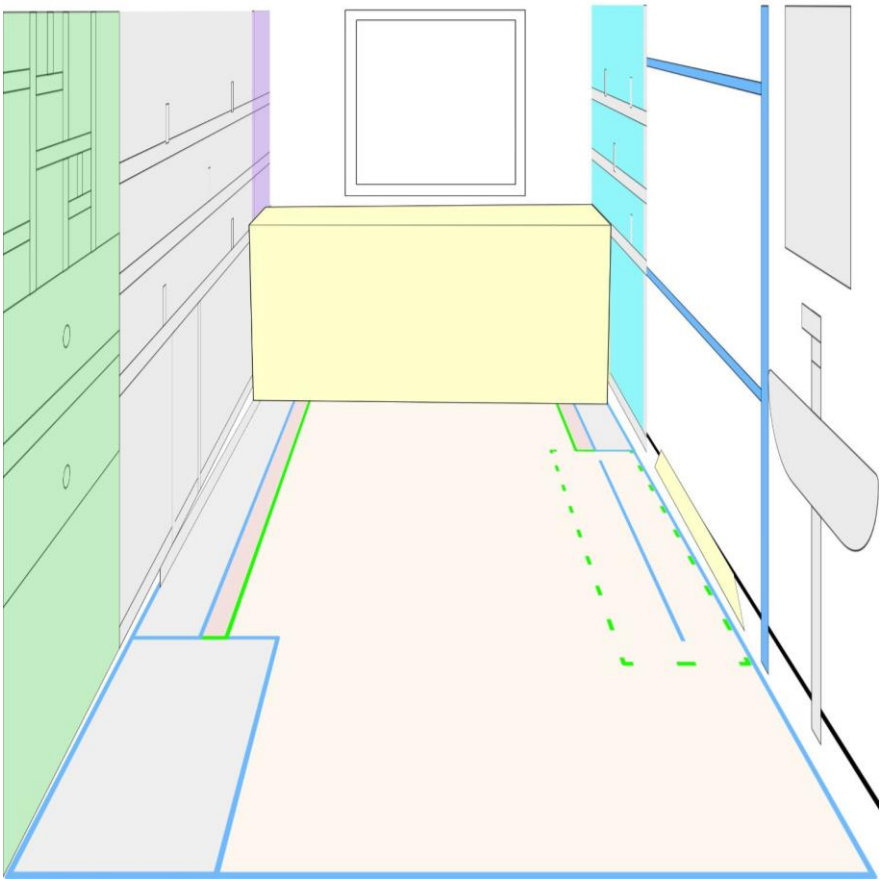


Figure 31: Final scenario layout visualization

Overall, the room tries to separate certain sections. And bring others closer, whilst simultaneously having plenty of *Open* storage to accommodate most objects. It also tries to have an accessible design that lets a lot of light into the room.

5.3 RePlace opening

As of the 24th of April 2026, the RePlace is open. It hasn't been long since the opening, but it seems to be running well thus far; knowledge of it has not yet spread widely. At this point, we are getting fewer users than the Återbruks loungen did. But it is also simply a quite new installation. There has been good volunteer interest. We do not, at this point, have access to the second room, but have been promised it will become available by the start of the next term.

As such, the RePlace is still in a sort of 'Beta' mode. And we know things will be moved around. The elaborate storage design imagined within this document has not materialised, but it still seems to be operating well.



Figure 32: 'Gulan' before RePlace



Figure 33: 'Gulan' during RePlace renovation



Figure 34: 'Gulan' after RePlace opening

As mentioned in point "4.1 Återbruks loungen", the major reason traditional Open shelving design can't be applied to spaces like Återbruks loungen is that, in traditional retail environments, the business is in control of how items are laid out, whereas in Återbruks loungen, users control where things go. In addition to the fact that retail environments have employees who actively clean up spaces. the RePlace has volunteers who clean up the space, and they also

exert social pressure to behave within the space. Hence, the RePlace is doing well without this kind of elaborate storage design.

6 Conclusion

I believe cumulative learning of this report could be useful in new types of sharing spaces, like the small-scale modular type. I believe they can be valuable. If a space is unmanned or excessively large, I think it can be helpful, as it grows and hopefully expands into more rooms, I believe the place will ultimately benefit more and more from this type of learning. I find sharing spaces exciting; having been involved for so long in them, I truly believe they can be part of creating change and providing meaningful non-capitalistic sharing. If only one takes the care to examine how they function.

This Design + Change project answers the research question: “How can the development of an organizing and storage system help create easily manageable self-service sharing spaces for a more circular economy?” as well as the sub-question: “How can sharing spaces become more efficient and user-friendly through their design, combatting the tragedy of the commons within the context of Linnaeus University?”

This project examined several types of sharing spaces to gain insight and find ways to answer the question “How can sharing spaces become more efficient and user-friendly? “

It did this through creating scenarios and looking at new ways and formats to create sharing spaces.

It also tries to accomplish this through the development of the concepts of *Open* versus *Closed,*

Collectability and *consumability.*

As well as with the development of the 9 point, as well as the concepts of open versus Closed shelving, and collectability and consumability in items. It has created tools for examining this type of shelf layout from a design perspective, hopefully allowing others to avoid the tragedy of the commons and to build upon these concepts to create new types of sharing spaces.

7 Bibliography

Arbell, Yael. 2023. ‘The Circle of Commoning: Conceptualising Commoning through the Case of Community-Led Housing.’ *International Journal of the Commons* 17 (1): 228–43. https://doi.org/10.5334/ijc.1210.

‘Asociación Forestal de Soria’. n.d. *The International Association for the Study of the Commons*. Accessed 13 April 2026. https://iasc-commons.org/cs-eoa-asfoso/.

Bergquist, M., P. Blumenschein, P. Karinti, et al. 2021. ‘Replicating the Focus Theory of Normative Conduct as Tested by Cialdini et al. (1990)’. *Journal of Environmental Psychology* 74 (April): 101573. https://doi.org/10.1016/j.jenvp.2021.101573.

‘Circularity Gap Report’. 2025. May. https://pdf.circularity-gap.world/?report=CGR_Global_2025_Report_0c90048033&page=1.

Creat. Commons (Creative Commons). n.d. ‘The Commons Belongs to Us All’. Accessed 13 April 2026. https://creativecommons.org/.

Frischmann, Brett M., Alain Marciano, and Giovanni Battista Ramello. 2019. ‘Retrospectives: Tragedy of the Commons after 50 Years’. *The Journal of Economic Perspectives* 33 (4): 211–28.

Gecili, Hakan, and Pratik J. Parikh. 2022. ‘Joint Shelf Design and Shelf Space Allocation Problem for Retailers’. *Omega (Oxford)* 111: 102634-. https://doi.org/10.1016/j.omega.2022.102634.

Gibson-Graham, J. K. 2013. *Take Back the Economy: An Ethical Guide for Transforming Our Communities*. With Jenny Cameron and Stephen Healy. University of Minnesota Press.

‘Gulan_planritning (1)’. n.d. Blueprint. Linnaeus University.

Hardin, Garrett. 1968. ‘The Tragedy of the Commons’. *Science* 162 (3859): 1243–48.

Jones, Natalie A., Helen Ross, Timothy Lynam, Pascal Perez, and Anne Leitch. 2011. ‘Mental Models: An Interdisciplinary Synthesis of Theory and Methods’. *Ecology and Society* 16 (1). https://www.jstor.org/stable/26268859.

Lundebye, A. 2004. ‘Senseness’. Masters, Goldsmiths, University of London.

Ostrom, Elinor, Joanna Burger, Christopher B. Field, Richard B. Norgaard, and David Policansky. 2017. ‘Revisiting the Commons: Local Lessons, Global Challenges’. In *Environment and Society*, edited by Christopher Schlottmann, Dale Jamieson, Colin

Jerolmack, Anne Rademacher, and Maria Damon. A Reader. NYU Press. https://doi.org/10.2307/j.ctt1ht4vw6.34.

‘Säg hej till TaGe!’ n.d. Text. Accessed 17 April 2026. https://kalmar.se/kommun-och-politik/hallbarhet-och-manskliga-rattigheter/ett-gronare-kalmar/sag-hej-till-tage.html.

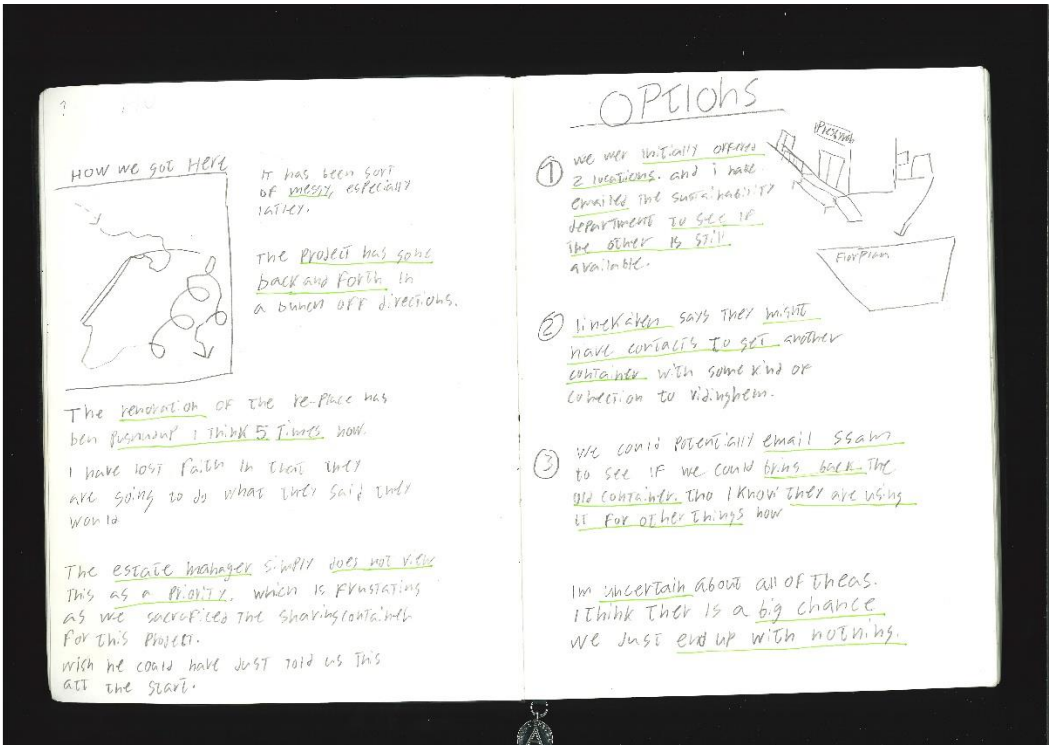
Tham, Mathilda. 2014. ‘Futures of Futures Studies in Fashion’. In Routledge Handbook of Sustainability and Fashion. Routledge.

Tham, Mathilda, Åsa Ståhl, and Sara Hyltén-Cavallius. 2019. Oikology - Home Ecologics : A Book about Building and Home Making for Permaculture and for Making Our Home Together on Earth. Linnaeus University Press. https://urn.kb.se/resolve?urn=urn:nbn:se:lnu:diva-90036.

Zajac, Sandra, and Heinrich Kuhn. 2025. ‘Joint Shelf Design and Space Planning Problem With Placement Options’. *Production and Operations Management*, July 7, 10591478251359808. https://doi.org/10.1177/10591478251359808.

Appendix:

Developmental sketches



HOW we got HERE

It has been sort of messy, especially lately.

The project has gone back and forth in a bunch OFF directions.

The renovation OF the re-Place has ben pushing I think 5 Times now.

I have lost faith in that they are going to do what they said they would

The estate manager simply does not view this as a PRIOrity. which is FRustating as we sacrificed the sharing container for this project.
Wish he could have just told us this all the start.

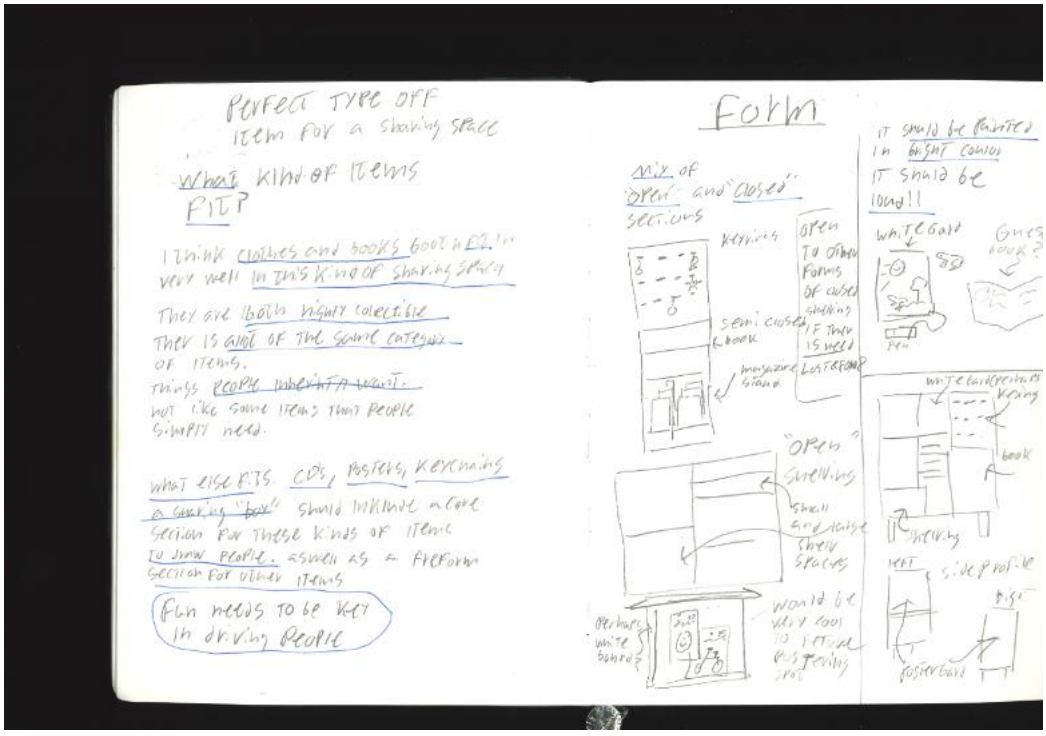
OPtIons

① we ver initially offered 2 locations. and i have emailed the sustainability department to see if the other is still available.

② linekaken says they might have contacts to get another container with some kind of connection to Ridinghem.

③ we could potentially email Ssam to see if we could bring back the old container. tho I know they are using it for other things now

Im uncertain about all of theas. I think ther is a big chance we just end up with nothing.



Perfect Type OFF Item For a Sharing Space

What kind OF Items FIT?

I think clothes and books both fit very well in this kind OF sharing space

They are both highly collectible
Ther is alot OF the same category OF Items.
Things PEOPLE inherit a want.
not like some items that people simply need.

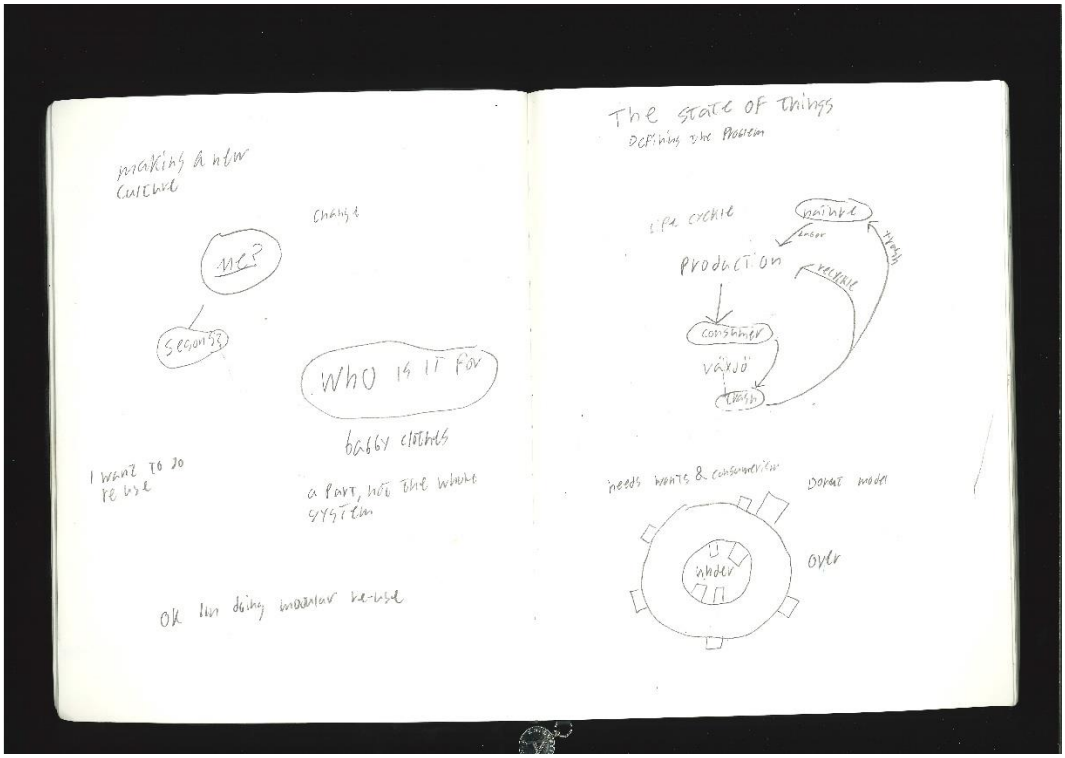
what else FITS. CDs, Posters, Keychains a Sewing "box" should indicate a core section For these kinds OF Items to show people. aswell as a FreeForm section For other items

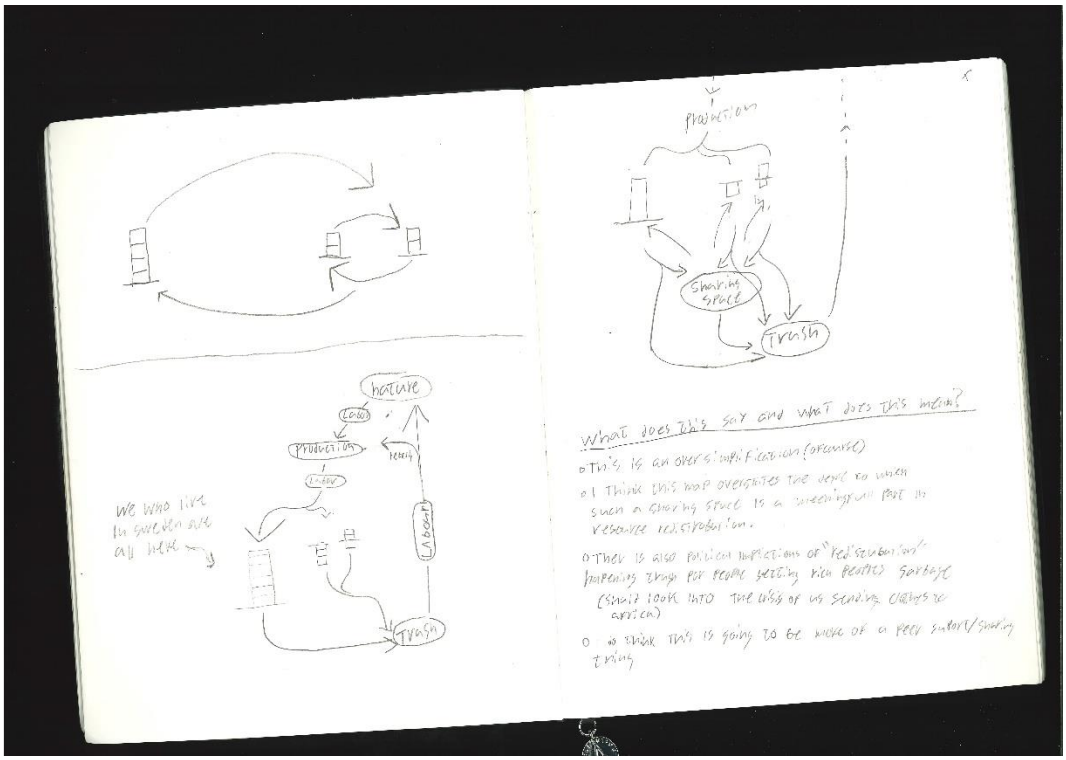
Fun needs to be key in driving People

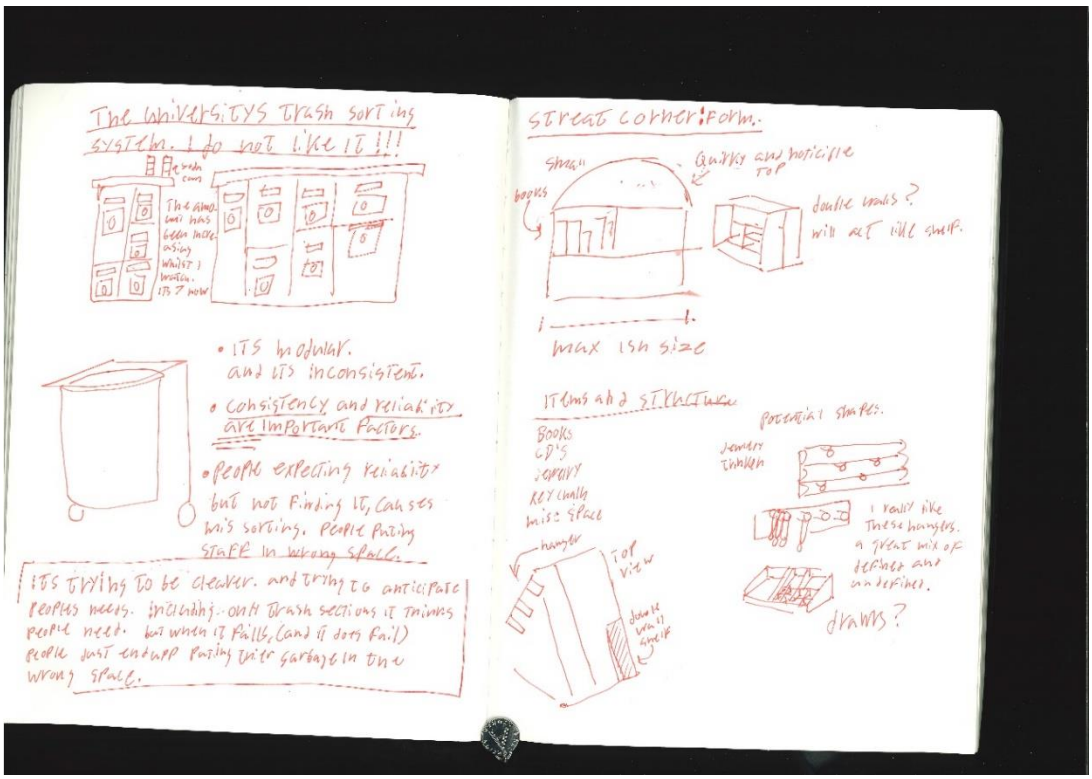
Form

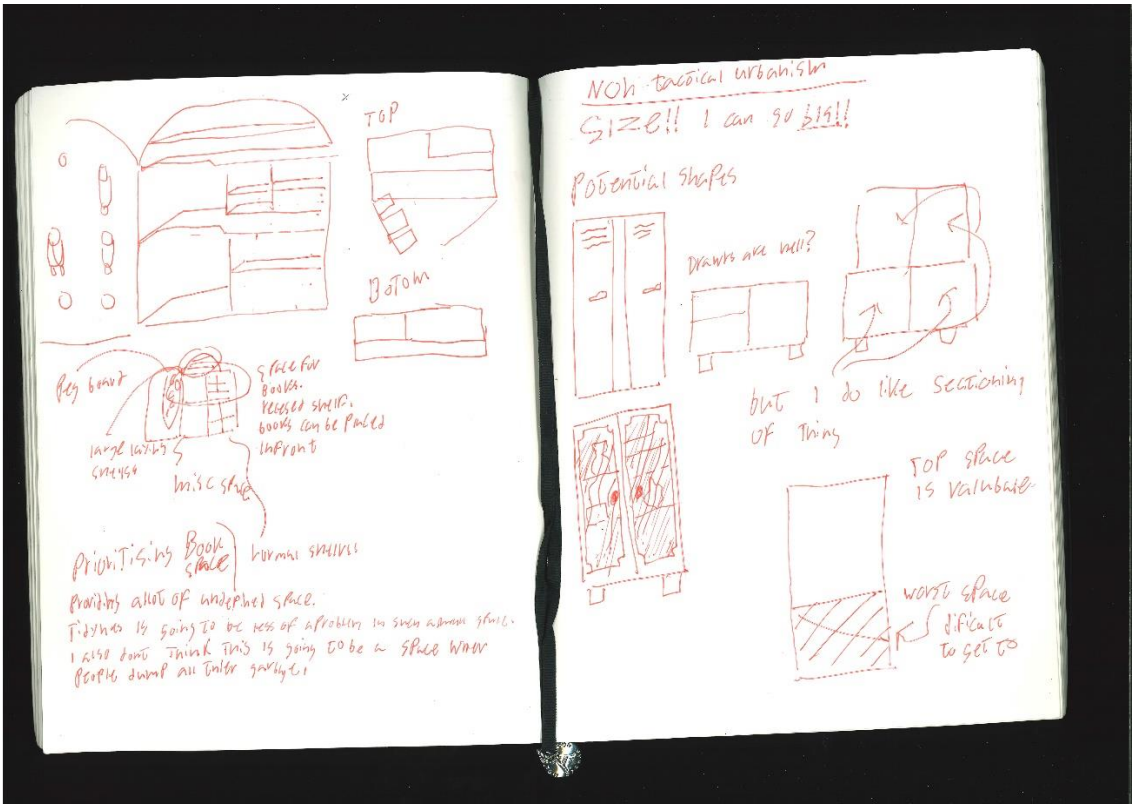
Mix OF "open" and "closed" sections

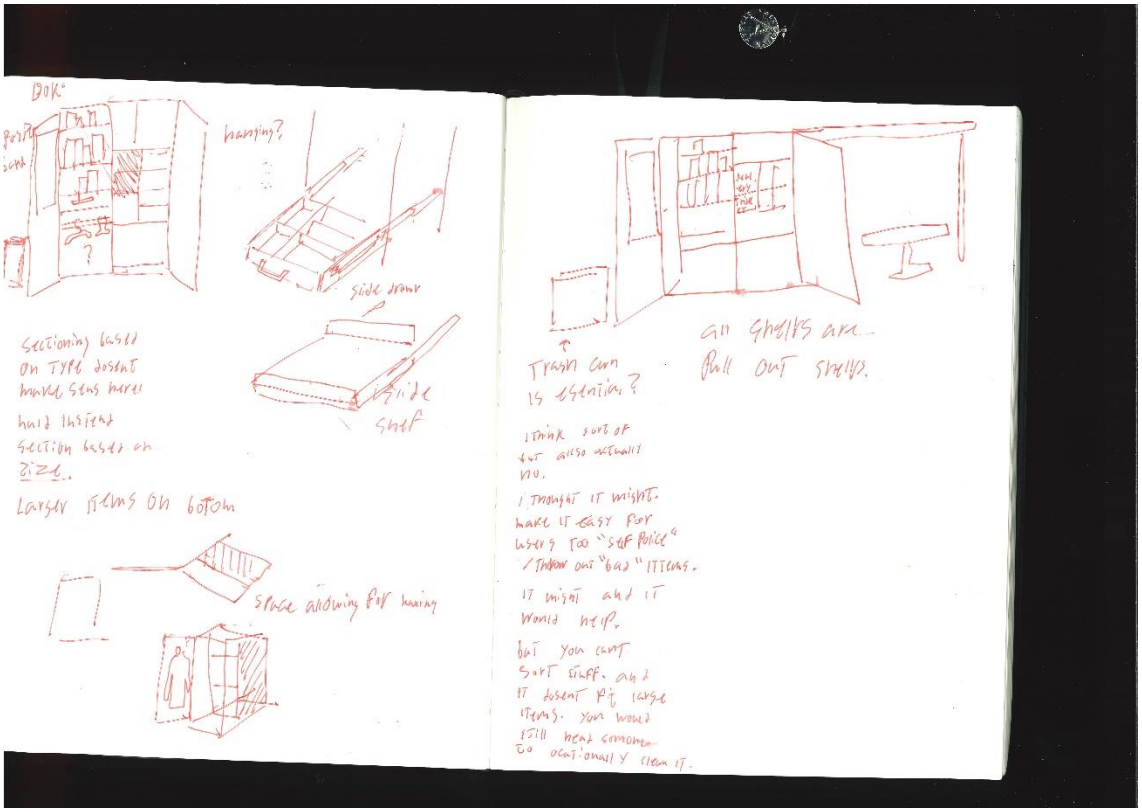
IT should be Painted in BRIGHT COLOUR IT should be loud!!

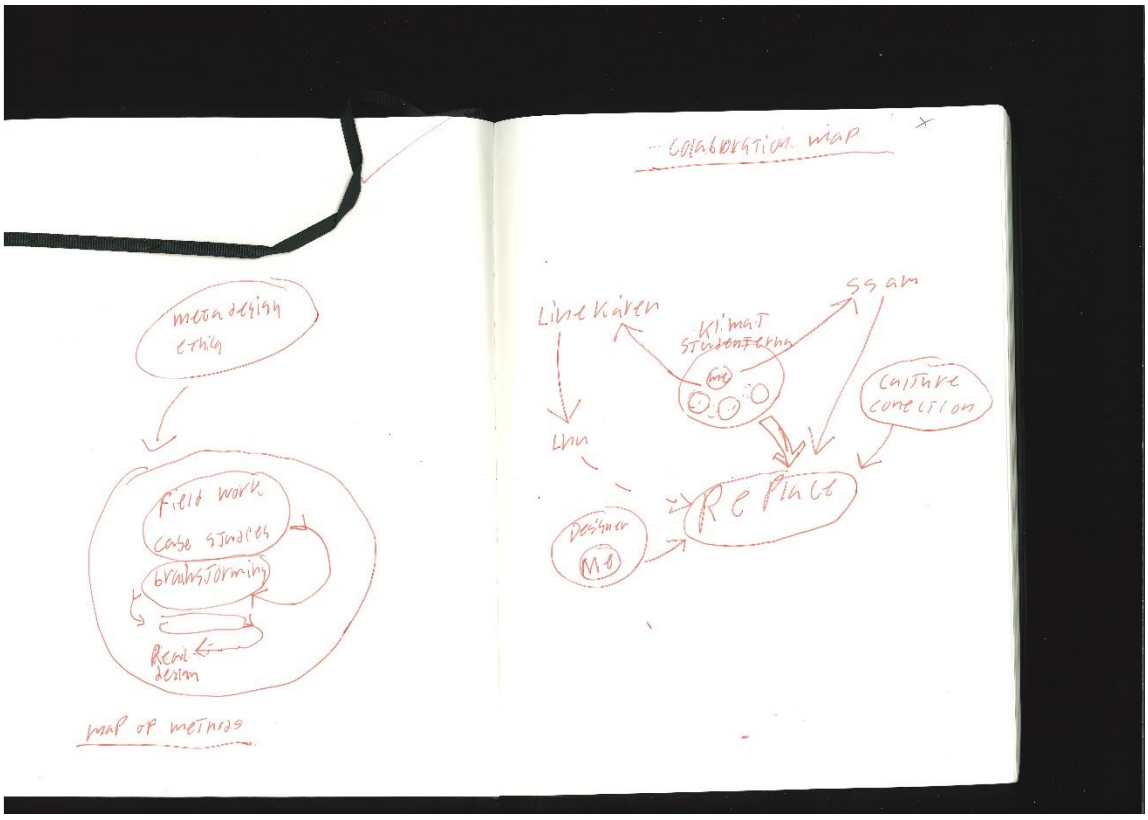

making a new current
Change
ME?
Secondly
WHO IS IT FOR
baggy clothes
I want to do re use
a part, not the whole system
OK I'm doing material re-use
The state of things
Defining the Problem
life cycle
material
Production
recycle
consumer
VÄRJÖ
trash
needs wants & consumption
donut model
under
over

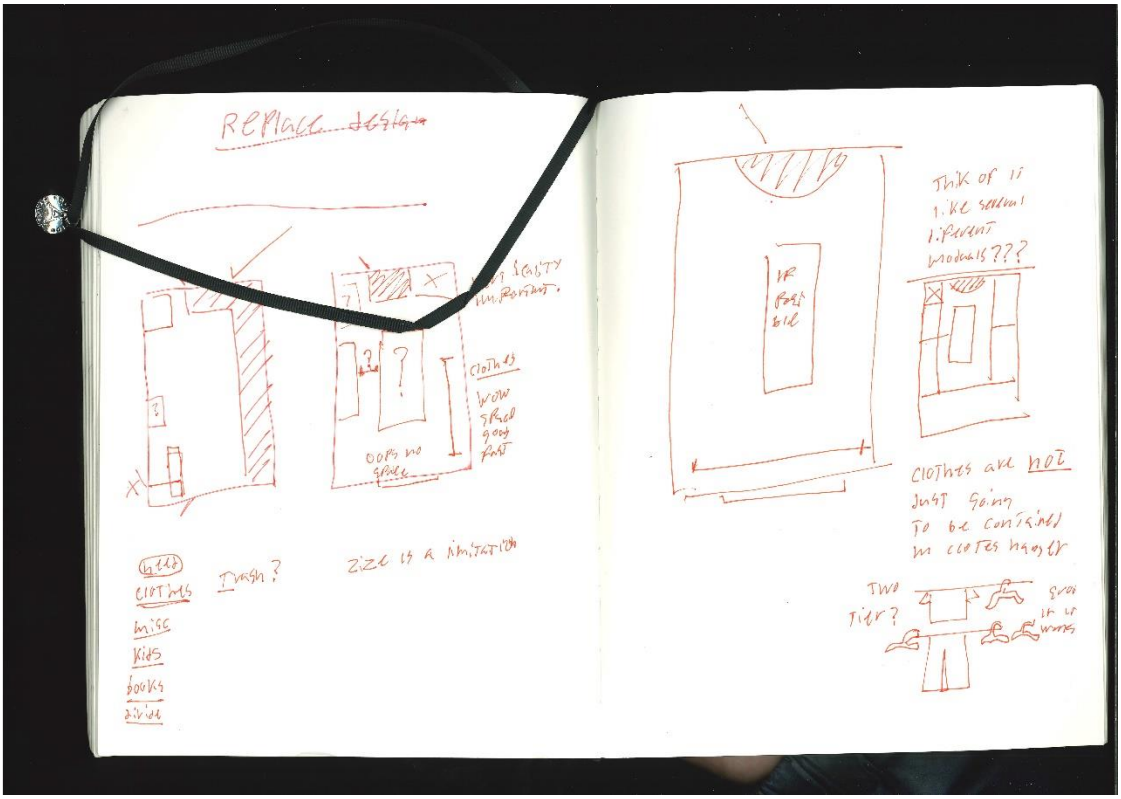

production
Sharing space
Trash
nature
LOSS
production
recycle
Labour
We who live in societies and all this
Trash
What does this say and what does this mean?
o This is an over simplification (obviously)
o I think this map overstates the ease to which such a sharing space is a sweeping out part in resource redistribution.
o There is also political implications of "peer to peer sharing" happening through for people selling their peoples garbage (should look into the use of us sending clothes to africa)
o so think this is going to be more of a peer sharing thing

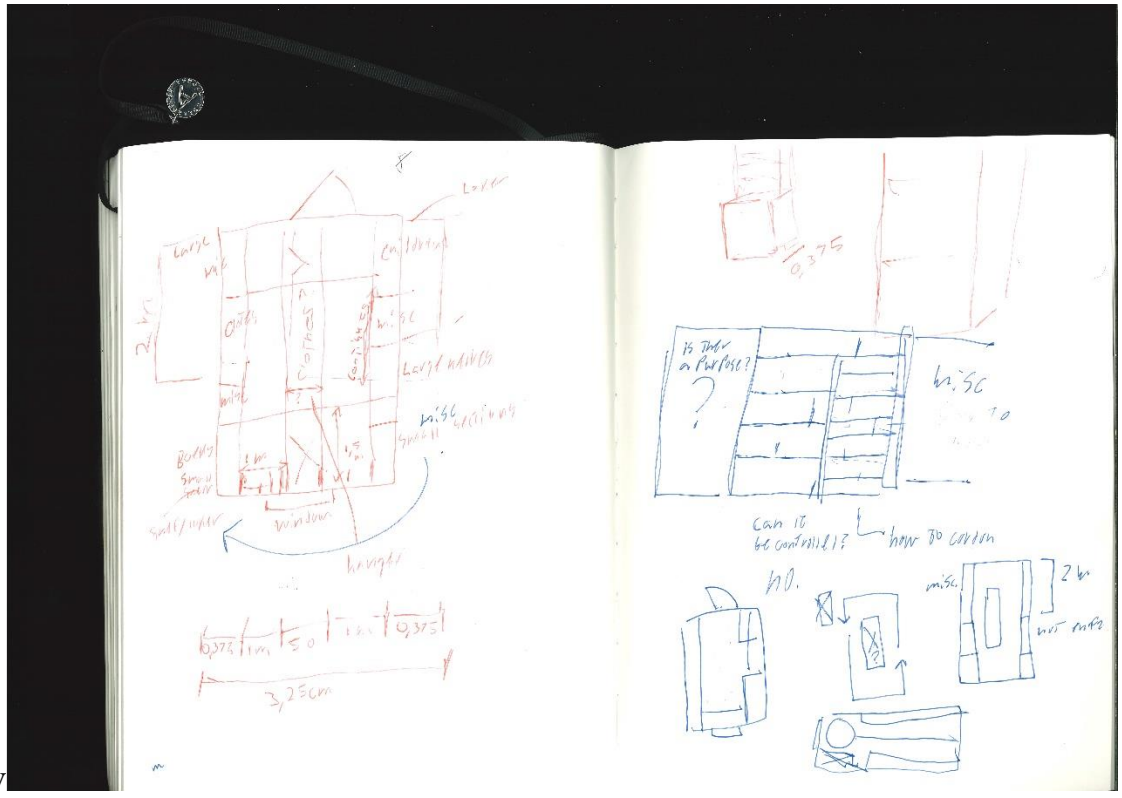

The University's trash sorting system. I do not like it!!!
trash cans
The aim of this has been made confusing whilst I write... it's now
• ITS modular. and its inconsistent.
• consistency and reliability are important factors.
• people expecting reliability but not finding it, causes mis sorting. people putting stuff in wrong space.
its trying to be clever. and trying to anticipate peoples needs. including own trash sections it thinks people need. but when it fails, (and it does fail) people just end up putting their garbage in the wrong space.
street corner: Form.
straight
books
Quirky and noticible top
double loads? will not fill shelf.
max ish size
Items and structure
Books
CD's
Stationary
Key cloths
misc space
hanger
top view
double unit shelf
potential shapes.
Jewelry Holders
I really like these hangers. a great mix of defined and undefined.
drawers?

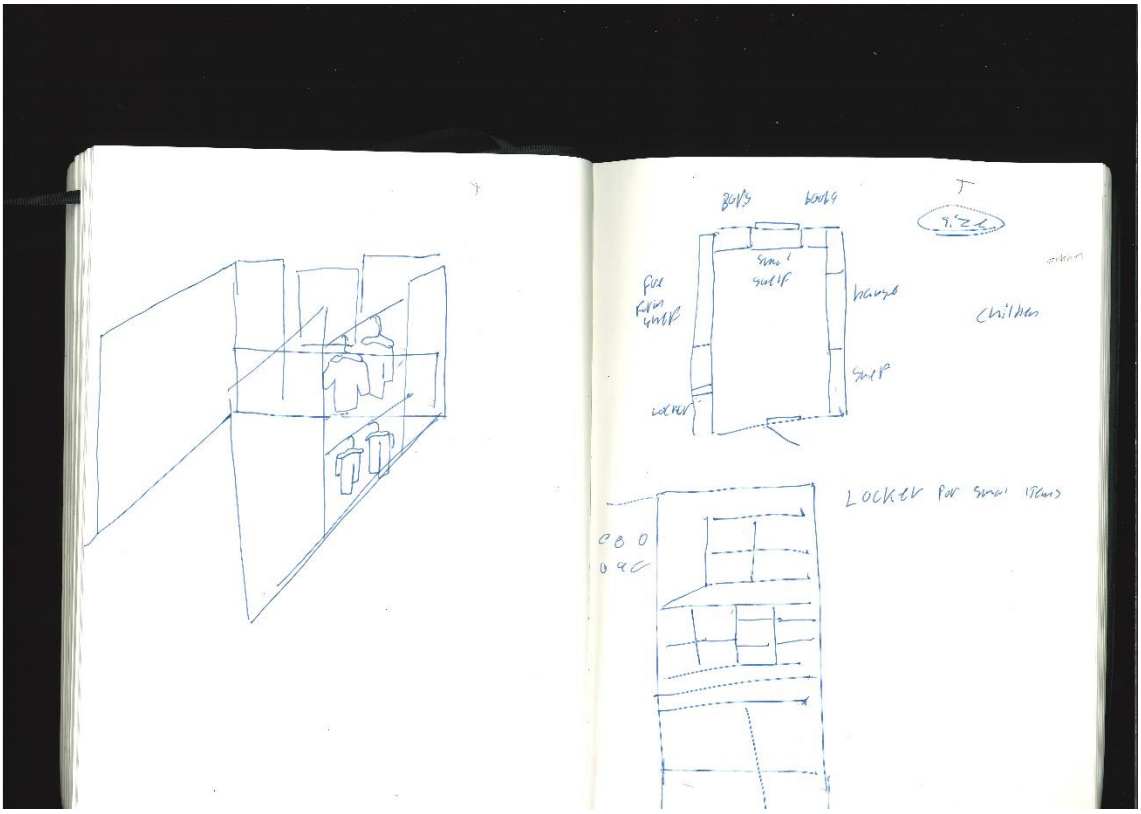

TOP
BOTTOM
Key board
large knick knacks
space for Books. recessed shelf. books can be placed infront
misc space
Prioritising Book space
Normal shelves
Providing allot of underused space.
Tidyness is going to be less of a problem in such a small space.
I also don't think this is going to be a space where people dump all their garbage.
NON tactical urbanism
Size!! I can go BIG!!
potential shapes
Drawers are tall?
but I do like sectioning of Thing
TOP space is valuable
worst space difficult to get to


DOK
hanging?
side drawer
Sectioning based on type doesnt make sens more!
hold this!
Section based on size.
Larger items on bottom
Space allowing for hanging
Trash can is station?
all shelfs are pull out shelfs.
I think sort of but also actually no.
I thought it might make it easy for users to "self police" / throw out "bad" items.
it might and it would help.
but you cant sort stuff. and it doesnt fit large items. you would still need someone to occationally clean it.


meta design ethics
field work
case studies
brainstorming
Real design
map of metrics
collaboration map
Linderiken
Klimat
Saab
Culture connection
LNU
Replace
Designer
MD




misc
Is there a purpose?
Can it be controlled?
how to control
misc
misc
misc


LOCKER for small items